DRUG AND COSMETIC ACT 1940 & RULES 1945

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DRUG & COSMETIC ACT, 1940

(This act was enacted on April 10, 1940)

Application of other laws not barred. —The provisions of this Act shall be in addition to, and not in derogation of, the Dangerous Drugs Act, 1930 (2 of 1930), and any other law for the time being in force.









Import

Regulates the import of drugs and cosmetics.

1

Manufacture

Regulates the manufacture of drugs and cosmetics.

2

Distribution

Regulates the distribution of drugs and cosmetics.

3

Sale

Regulates the sale of drugs and cosmetics.

4

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CHAPTER 1: INTRODUCTORY

DEFINITIONS

1. Drug

- All medicines for internal or external use of human beings or animals and all substances intended to be used for or in the diagnosis, treatment, mitigation or prevention of any disease or disorder in human beings or animals, including preparations applied on human body for the purpose of repelling insects like mosquitoes.
- Such substances (other than food) intended to affect the structure or any function of the human body or intended to be used for the destruction of [vermin] or insects which cause disease in human beings or animals, as may be specified from time to time by the Central Government by notification in the Official Gazette.
- All substances intended for use as components of a drug including empty gelatin capsules; and
- Such devices intended for internal or external use in the diagnosis, treatment, mitigation or prevention of disease or disorder in human beings or animals, as may be specified from time to time by the Central Government by notification in the Official Gazette, after consultation with the Board.

CHAPTER 1: INTRODUCTORY

DEFINITIONS

2. Ayurvedic, Siddha or Unani Drug

- Includes all medicines intended for internal or external use for or in the diagnosis, treatment, mitigation or prevention of disease or disorder in human beings or animals and manufactured exclusively in accordance with the formulae described in, the authoritative books of Ayurvedic, Siddha and Unani Tibb system of medicine, specified in the First Schedule.

3. Cosmetic

- Means any article intended to be rubbed, poured, sprinkled or sprayed on, or introduced into, or otherwise applied to, the human body or any part thereof for cleansing, beautifying, promoting attractiveness, or altering the appearance, and includes any article intended for use as a component of cosmetic.

4. Board

- In relation to Ayurvedic, Siddha or Unani drug, the Ayurvedic, Siddha and Unani Drugs Technical Advisory Board constituted under section 33C; and in relation to any other drug or cosmetic, the Drugs Technical Advisory Board constituted under section 5.

CHAPTER 1: INTRODUCTORY

DEFINITIONS

5. Government Analyst

- In relation to Ayurvedic, Siddha or Unani drug, a Government Analyst appointed by the Central Government or a State Government under section 33F; and in relation to any other drug or cosmetic, a Government Analyst appointed by the Central Government or a State Government under section 20.

6. Inspector

- In relation to Ayurvedic, Siddha or Unani drug, an Inspector appointed by the Central Government or a State Government under section 33G; and in relation to any other drug or cosmetic, an Inspector appointed by the Central Government or a State Government under section 21.

7. Manufacture

- In relation to any drug or cosmetic includes any process or part of a process for making, altering, ornamenting, finishing, packing, labeling, breaking up or otherwise treating or adopting any drug or cosmetic with a view to its sale or distribution but does not include the compounding or dispensing of any drug, or the packing of any drug or cosmetic, in the ordinary course of retail business; and “to manufacture” shall be construed accordingly.

CHAPTER 1: INTRODUCTORY

DEFINITIONS

8. Import

- Means to bring into India.

9. Patent or Proprietary Medicine

- In relation to Ayurvedic, Siddha or Unani Tibb systems of medicine all formulations containing only such ingredients mentioned in the formulae described in the authoritative books of Ayurvedic, Siddha or Unani Tibb systems of medicine specified in the First Schedule but does not include a medicine which is administered by parenteral route and also a formulation included in the authoritative books as specified in clause (a).
- In relation to any other systems of medicine, a drug which is a remedy or prescription presented in a form ready for internal or external administration of human beings or animals and which is not included in the edition of the Indian Pharmacopoeia for the time being or any other Pharmacopoeia authorized in this behalf by the Central Government after consultation with the Drugs Technical Advisory Board constituted under section 5.

10. Presumption as to poisonous substances

- Any substance specified as poisonous by rule made under Chapter II or Chapter IV or Chapter IVA shall be deemed to be a poisonous substance.

CHAPTER 1: INTRODUCTORY

Schedules to Act

First Schedule: Names of books under Ayurvedic, Siddha and Unani Tibb systems.
Second Schedule: Standard to be compiled with by imported drugs and by drugs manufactured for sale, stocked or exhibited for sale, sold or distributed.

In addition, the following appendices are also prescribed:
Appendix I: Data required to be submitted with application for permission to market a new drug.
Appendix II: Format for submission of clinical trial reports.
Appendix III: Animal toxicity requirements for clinical trials and marketing of a new drug.
Appendix IV: Number of animals for long term toxicity studies.
Appendix V: Patient consent form participation in a Phase I clinical trial.
Appendix VI: Four groups of fixed dose combinations and their data requirements.

CHAPTER 1: INTRODUCTORY

Schedules to Rules

Schedule A: Contains various forms and formats of letters for applications of licensing etc.
Schedule B: Contains fees structure for government-run labs.
Schedule C: Contains various biological products and their regulation. Examples: serums, adrenaline, vitamins etc.
Schedule D: List of drugs exempted from the provision of import of drugs
Schedule E: Contains various poisons and their regulation. Examples: Sarpa Visha (Snake venom), Parada (Mercury) etc.
Schedule F: This contains regulations and standards for running a blood bank.
Schedule F-I: This contains regulations and standards for vaccines.
Schedule F-II: This contains regulations and standards for surgical dressing.
Schedule F-III: This contains regulations and standards for umbilical tapes.
Schedule FF: This contains regulations and standards for ophthalmic ointments and solutions.
Schedule G: These drugs include hormonal medications (excluding sex hormone medications), antineoplastic drugs, anticonvulsants, hypoglycemic drugs, antihistamines, etc. The drug label must display the text "*Caution: It is dangerous to take this preparation except under medical supervision*" prominently. Examples of substances under this schedule: All insulin preparations, hydroxyurea, daunorubicin, metformin.

CHAPTER 1: INTRODUCTORY

Schedules to Rules

Schedule H: Each drug's label must prominently display the symbol "**Rx**" and a red-boxed warning "Schedule H drug. **Warning : "Not to be sold by Retail without the prescription of a Registered Medical Practitioner**". It can only be supplied to licensed parties. It cannot be sold without a prescription and only the amount specified in the prescription should be sold. The time and date of prescription must be noted. Examples: Alprazolam (Xanax), Hepatitis B vaccine, Ibuprofen, Vasopressin etc.

If a Schedule H drug also comes under the purview of Narcotic Drugs and Psychotropic Substances Act, 1985, the drug's label must prominently display the symbol "**NRx**" and a red-boxed warning "Schedule H drug. Warning : "**Not to be sold by Retail without the prescription of a Registered Medical Practitioner**".

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Schedule H1: Notified in 2013, this list includes third and fourth generation antibiotics, some psychotropic drugs and anti-TB drugs. A separate register is to be maintained to track supply of these drugs and labelling requirements are of the symbol "**Rx**" and the red boxed warning : "Schedule H1 Drug-Warning: It is dangerous to take this preparation except in accordance with the medical advice. -Not to be sold by retail without the prescription of a Registered Medical Practitioner."

Schedule J: Contains a list of various diseases and medical conditions that cannot be treated under any drug currently in market. No drug may legally claim to treat these diseases

CHAPTER 1: INTRODUCTORY

Schedules to Rules

Schedule K: Drugs not meant for medicinal use, quinine and other antimalarial drugs, drugs supplied by government hospitals, registered medical practitioners, contraceptive drugs, and their corresponding regulation.
Schedule M: Contains various regulations for manufacturing, premises, waste disposal and equipment.
Schedule N: Contains various regulations and requirements for a pharmacy
Schedule O: Contains various regulations and requirements for disinfectant fluids.
Schedule P: Contains regulations regarding life period and storage of various drugs.
Schedule P-I: Contains regulations regarding retail package size of various drugs.
Schedule Q: Contains a list of permitted dyes and pigments in soap and cosmetics.
Schedule R: Contains various regulations and requirements for condoms and other mechanical contraceptives.
Schedule S: Lists various cosmetics and toiletries and directs the manufacturers of cosmetics to conform to the latest Bureau of Indian Standards requirements.
Schedule T: Contains various regulations and requirements for manufacture of Ayurvedic, Siddha and Unani drugs.
Schedule U: Contains various regulations and requirements for record keeping.
Schedule V: Contains standards for drug patents.

CHAPTER 1: INTRODUCTORY

Schedules to Rules

Schedule W: Lists generic drugs.
Schedule X: Schedule X lists addictive drugs (e.g. narcotics, psychotropics) having medicinal uses that must be kept under lock and key. All the regulations of Schedule H apply. The retailer must keep a copy of the prescription for two years. Various licenses are required for possession and storage of these drugs, including a special license from the local drug controller. Examples: morphine, secobarbital, glutethimide etc.
Schedule Y: Contains requirement and guidelines for clinical trials.
Schedule Z: Lists approved drugs for veterinary use.

CHAPTER II: THE DRUGS TECHNICAL ADVISORY BOARD, THE CENTRAL DRUGS LABORATORY AND THE DRUGS CONSULTATIVE COMMITTEE

Administrative Bodies

- **Advisory**
  - Drug Technical Advisory Board (DTAB)
  - Drug Consultative Committee (DCC)
- **Analytical**
  - The Central Drug Laboratory (CDL)
  - Drug Control Laboratories in the state (DCL)
  - Government Analyst
- **Executive**
  - Licensing Authorities (central & state)
  - Drug Inspectors
  - Custom Collector

CHAPTER II: THE DRUGS TECHNICAL ADVISORY BOARD, THE CENTRAL DRUGS LABORATORY AND THE DRUGS CONSULTATIVE COMMITTEE

DTAB (The Drugs Technical Advisory Board)

The Central Government shall, as soon as may be, constitute a Board to advise the Central Government and the State Governments on technical matters arising out of the administration of this Act and to carry out the other functions assigned to it by this Act.

The Board shall consist of the following members, namely:

(i) the Director General of Health Services, ex officio, who shall be the Chairman
(ii) the Drugs Controller, India, ex officio
(iii) the Director of the Central Drugs Laboratory, Calcutta, ex officio
(iv) the Director of the Central Research Institute, Kasauli, ex officio
(v) the Director of the Indian Veterinary Research Institute, Izatnagar, ex officio
(vi) the President of the Medical Council of India, ex officio
(vii) the President of the Pharmacy Council of India, ex officio
(viii) the Director of the Central Drug Research Institute, Lucknow, ex officio
(ix) two persons to be nominated by the Central Government from among persons who are in charge of drugs control in the States.
(x) one person, to be elected by the Executive Committee of the Pharmacy Council of India, from among teachers in pharmacy or pharmaceutical chemistry or pharmacognosy on the staff of an Indian university or a college affiliated thereto
(xi) one person, to be elected by the Executive Committee of the Medical Council of India, from among teachers in medicine or therapeutics on the staff of an Indian university or a college affiliated thereto.

CHAPTER II: THE DRUGS TECHNICAL ADVISORY BOARD, THE CENTRAL DRUGS LABORATORY AND THE DRUGS CONSULTATIVE COMMITTEE

(xii) One person to be nominated by the Central Government from the pharmaceutical industry
(xiii) One pharmacologist to be elected by the Governing Body of the Indian Council of Medical Research.
(xiv)One person to be elected by the Central Council of the Indian Medical Association.
(xv) One person to be elected by the Council of the Indian Pharmaceutical Association
(xvi)Two persons holding the appointment of Government Analyst under this Act, to be nominated by the Central Government.

Points to Ponder

1. The nominated and elected members of the Board shall hold office for three years, but shall be eligible for re-nomination and re-election provided that the person nominated or elected under clause (ix) or clause (x) or clause (xi) or clause (xvi) of sub-section (2) shall hold office for so long as he holds the appointment of the office by virtue of which he was nominated or elected to the Board.
2. The Board may, subject to the previous approval of the Central Government, make bye-laws fixing a quorum and regulating its own procedure and the conduct of all business to be transacted by it.
3. The Board may constitute sub-committees and may appoint to such sub-committees for such periods, not exceeding three years, as it may decide, or temporarily for the consideration of particular matters, persons who are not members of the Board.
4. The functions of the Board may be exercised notwithstanding any vacancy therein.
5. The Central Government shall appoint a person to be Secretary of the Board and shall provide the Board with such clerical and other staff as the Central Government considers necessary.

CHAPTER II: THE DRUGS TECHNICAL ADVISORY BOARD, THE CENTRAL DRUGS LABORATORY AND THE DRUGS CONSULTATIVE COMMITTEE

Products	Lab
sera, vaccines, toxins, antigen, antitoxins, sterilized surgical sutures & ligatures etc.	Central Research Institute, Kasauli
sera, vaccines, toxins, antigen, antitoxins for veterinary use	Indian Veterinary Research Institute, Izzatnagar
Oral Polio vaccines	• Vaccine testing laboratory of CRI kasuali. • National Inst. Of Communicable diseases • National Inst. Of Vaccines, Pune
VDRL Antigen	Laboratory of Serologist & Chemical Examiner to the Govt. of India
Condoms	Indian Pharmacopoeial Laboratory, Gaziabad

CDL (The Central Drugs Laboratory)

- The Central Government shall, as soon as may be, establish a Central Drugs Laboratory under the control of a Director to be appointed by the Central Government, to carry out the functions entrusted to it by this Act or any rules made under this Chapter.
- Provided that, if the Central Government so prescribes, the functions of the Central Drugs Laboratory in respect of any drug or class of drugs [or cosmetic or class of cosmetics] shall be carried out at the Central Research Institute, Kasauli, or at any other prescribed Laboratory and the functions of the Director of the Central Drugs Laboratory in respect of such drug or class of drugs [or such cosmetic or class of cosmetics] shall be exercised by the Director of that Institute or of that other Laboratory, as the case may be.
- Functions
  - To analyze the samples of drugs or cosmetics sent to it by **(a)** Custom collector or any authorized officer **(b)** by court
  - To carry out such other duties as may be assigned to it by central govt. or state govt. with the permission of central govt. after consultation with DTAB.
  - In case of some drugs like biological and microbiological tests are not carried out in CDL at Kolkata hence the test or analysis of such samples or functions of CDL is carried out at other labs or institutions.

CHAPTER II: THE DRUGS TECHNICAL ADVISORY BOARD, THE CENTRAL DRUGS LABORATORY AND THE DRUGS CONSULTATIVE COMMITTEE

The Drugs Consultative Committee (DCC)

1. The Central Government may constitute an advisory committee to be called “the Drugs Consultative Committee” to advise the Central Government, the State Governments and the Drugs Technical Advisory Board on any matter tending to secure uniformity throughout India in the administration of this Act.
2. The Drugs Consultative Committee shall consist of two representatives of the Central Government to be nominated by that Government and one representative of each State Government to be nominated by the State Government concerned.
3. The Drugs Consultative Committee shall meet when required to do so by the Central Government and shall have power to regulate its own procedure.

Sections 5 and 7 not to apply to Ayurvedic, Siddha or Unani drugs.—Nothing contained in sections 5 and 7 shall apply to Ayurvedic, Siddha or Unani drugs.

Government Analyst

The **Central & State Government**, by notification in the Official Gazette, may appoint persons having the prescribed qualifications as Government Analysts for:

- specified areas in the State; and
- specified drugs or classes of drugs;
- specified cosmetics or classes of cosmetics.

Qualifications

Route	Educational qualification	Experience/training
A	Graduate in Medicine/Science/Pharmacy/Pharmaceutical Chemistry	≥5 years post-graduate experience in drug testing **OR** specified CDL training
B	PG degree in Medicine/Science/Pharmacy/Pharmaceutical Chemistry **OR** Associateship Diploma of Institution of Chemists (India) with Analysis of Drugs & Pharmaceuticals	≥3 years experience in drug testing / specified training
Schedule C	Additional requirements depending on educational background	6 months, 3 years or specified CDL training
Veterinary biologicals	Graduate in relevant discipline	≥5 years biological-product standardization
Veterinary biologicals: PG route	PG in relevant discipline	≥3 years biological-product standardization
Conflict of interest	—	Cannot be directly/indirectly engaged in drug manufacturing business

Duties of Government Analyst

1. Analyse/test samples

The Government Analyst shall cause to be **analysed or tested** samples of drugs, and cosmetics sent by Inspectors, or other persons authorized under Chapter IV of the Act.

2. Furnish analytical reports

After testing/analysis, the Government Analyst shall furnish reports of the results in accordance with the Rules.

3. Submit analytical/research reports to Government

The Government Analyst shall periodically forward to the Government reports containing results of analytical work, and research. These may be considered for publication at the discretion of the Government.

Procedure when sample is received

When a package containing a sample is received from an Inspector, the Government Analyst must first; (1) **Examine the seal** on the packet, or the portion of sample/container with the **specimen impression of the seal** received separately.

(2) Record condition of seal; The condition of the seal on the packet/sample/container must be noted to establishe the **integrity of the sample**.

Before testing, the analyst should be satisfied that the sample received corresponds to the sealed sample submitted by the Inspector.

After completing the test or analysis, the Government Analyst must provide the Inspector with a report, **in triplicate,** in **Form 13**, together with the required protocols of the tests/analysis applied.

Form 13 is the **"Certificate of test or analysis by Government Analyst.** The form contains:

1. Name of Inspector from whom sample was received
2. Serial number and date of Inspector's memorandum
3. Number of sample
4. Date of receipt

5. Name of drug purported to be contained in sample
6. Condition of seals
7. Result of test/analysis and protocols applied
8. Opinion whether the sample is of **standard quality or not of standard quality**
9. Reasons supporting the opinion
10. Date and signature of Government Analyst

Protocols of tests/analysis

A. Pharmacopoeial drug or Patent/ Proprietary Drug: If the official pharmacopoeial test/method is followed, the report should give a **reference to the specific pharmacopoeial test or analysis**.

B. Modified pharmacopoeial method: If a pharmacopoeial drug is tested using a **modified official test/method,** the report should describe the **actual test/assay method applied**.

C. No pharmacopoeial method available: If the analyst follows methods from standard books, or scientific journals, the report should include:

- description of the test/method used, and
- reference to the relevant book/journal.

D. Method developed by Government Analyst: Where no test method is available and the Government Analyst develops a method, the **description of the test applied** must be included in the report.

Distribution of Government Analyst's report

Copy	Recipient / purpose
1st copy	Person from whom sample was taken
2nd copy	Person whose name/address/particulars were disclosed under Section 18A, where applicable
3rd copy	Retained for use in prosecution

If the Government Analyst's report is challenged?

If the concerned person gives the required notice within 28 days, the Court may send the sample produced before the Magistrate to the **Central Drugs Laboratory** for further testing/analysis.

The Central Drugs Laboratory then prepares a written report:

- signed by, or
- under the authority of,

the **Director of the Central Drugs Laboratory**.

The report is treated as **conclusive evidence of the facts stated in it**.

The cost of the test/analysis carried out by the Central Drugs Laboratory is paid by the complainant, or the accused, as directed by the Court.

❖ A person, or a **recognized consumer association**, can submit a purchased drug, or cosmetic to a Government Analyst for test or analysis.

The person/association must:

1. make the application in the prescribed manner;
2. pay the prescribed fee;
3. submit the purchased drug/cosmetic.

The applicant is entitled to receive a report signed by the Government Analyst.

Form 14A and Form 14B

Form 14A: Used for the **application by a purchaser for test or analysis.**

The application includes details such as:

- full name and address of applicant;
- occupation;
- name of drug;
- name/address of pharmacy or concern from which purchased;
- date of purchase;
- reason for submitting the drug for testing;
- prescribed fee details;
- declaration that the sample was purchased by/for the applicant and has not been tampered with.

Form 14B: Used for the **Certificate of test or analysis by Government Analyst.**

It includes:

- name of person from whom sample was received;
- date of receipt;
- name of drug;
- opinion whether the sample is or is not of standard quality;
- date and signature of Government Analyst.

DRUG INSPECTORS

State or Central Government → appoints qualified person → assigns area → Inspector exercises prescribed powers and performs prescribed duties.

A person having financial interest in import, manufacture or sale of drugs/cosmetics cannot be appointed as Inspector. Inspector is treated as a public servant

Qualifications

A. Basic academic qualification

- Degree in Pharmacy
- Degree in Pharmaceutical Sciences
- Degree in Medicine with specialization in Clinical Pharmacology
- Degree in Medicine with specialization in Microbiology

*The degree must be from a **University established in India by law**.*

Additional requirement for inspection of Schedule C manufacture

Route	Minimum requirement
Manufacturing experience	**18 months** experience in manufacture of at least one Schedule C substance
Testing experience	**18 months** experience in testing at least one Schedule C substance in a laboratory approved by the Licensing Authority
Inspection experience	**3 years** experience in inspection of firms manufacturing Schedule C substances during service as Drug Inspector

Duties of Inspectors

Duty	Explanation
Inspect licensed establishments	Inspect all establishments licensed for sale of drugs in the assigned area
Frequency of inspection	**Not less than once a year**
Check licence conditions	Ensure conditions of the licence are being followed
Suspected imported packages	Procure and send suspected packages for test/analysis when necessary
Investigate complaints	Investigate complaints made **in writing**
Institute prosecution	Take prosecution action for breaches of the Act/Rules
Maintain inspection records	Record inspections and actions taken
Record sampling/seizure	Include details of samples taken and stocks seized
Submit records	Submit copies of records to controlling authority

Detect illegal sale	Conduct enquiries and inspections necessary to detect contravention
Detain imported packages	When authorized by State Government, detain imported packages suspected to contain prohibited drugs

Powers of Drug Inspector

Power	What the Inspector can do
Inspect manufacturing premises	Inspect premises where drugs/cosmetics are manufactured
Inspect testing/standardization facilities	Examine means employed for standardization and testing
Inspect sale premises	Inspect places where drugs/cosmetics are sold, stocked, exhibited, offered for sale or distributed
Take samples	Take samples of drugs/cosmetics
Search persons	Search a person when there is reason to believe that an offending drug/cosmetic is secretly kept on his person
Enter premises	Enter and search premises when there is reason to believe an offence has been or is being committed
Search conveyances	Stop and search vehicles, vessels or other conveyances suspected of carrying offending drugs/cosmetics
Examine documents	Examine records, registers, documents and other relevant material
Prohibit disposal	Order a person not to dispose of specified stock
Seize stock	Seize drugs/cosmetics or relevant substances/articles in specified circumstances
Seize records/material	Seize relevant records, registers, documents or material objects as permitted
Detain imported packages	In applicable circumstances and with required authorization

Prohibition of disposal ≠ seizure?

Prohibition of disposal	Seizure
Stock remains with the person	Stock/material is taken into official custody
Person is ordered not to sell/dispose	Inspector physically seizes the material
Form 15	Receipt in Form 16 for seized material

Obstruction of Inspector

Situation	Consequence stated in the Act
Wilfully obstructing Inspector	Punishable
Refusing to produce required record/register/document	Punishable
Maximum imprisonment stated in the uploaded provision	**Up to 3 years**
Fine	May also be imposed
Both	Imprisonment and fine may apply

Procedure for taking samples

- Decide to take sample
- Tender **fair price** of sample
- Obtain written acknowledgment, if required
- If price is refused, issue prescribed receipt
- Inform person in writing about purpose of sampling
- Divide sample into prescribed portions / take required containers
- Seal and mark the sample
- Permit person to add his own seal/mark
- Give one portion to person from whom sample is taken
- Send appropriate portion to Government Analyst
- Retain portion for Court, where applicable
- Send additional portion to manufacturer/person disclosed under Section 18A, where applicable
- Dispatch Analyst's portion according to Rule 57

Division of sample

Situation	Number/arrangement
Normal sampling	Sample divided into **four portions**
Sample from manufacturing premises	**Three portions**
Small-volume containers	Appropriate number of containers taken instead of ordinary division, as prescribed
Sealing	Inspector effectively seals the sample
Marking	Sample is suitably marked
Additional seal	Person from whom sample is taken may add his own seal/mark

Distribution of Sample Portions

One portion	Returned to person from whom sample was taken
One portion	Sent to Government Analyst
One portion	Used for Court proceedings, if instituted
Additional portion	Sent to manufacturer/person whose particulars are disclosed under Section 18A, where applicable

Controlling Authority

The **Controlling Authority** is the officer appointed by the Central Government or State Government to exercise control over the Drug Inspectors appointed by that Government.

Qualification of Controlling Authority

Requirement	Qualification
Academic qualification	Graduate in **Pharmacy** OR **Pharmaceutical Chemistry** OR **Medicine with specialization in Clinical Pharmacology or Microbiology**
University	Degree should be from a University established in India by law
Experience	Must have experience in the **manufacture or testing of drugs** or in **enforcement of the provisions of the Act**
Purpose	Ensures that the person supervising Drug Inspectors has adequate technical and regulatory knowledge

Role of the Controlling Authority in Drug Inspector Work

1. At sale premises

Inspector's activity	Role of Controlling Authority
Inspection of licensed establishments	Provides supervisory control/instructions
Investigation of complaints	Supervises official action
Taking of samples	Receives/maintains inspection records as prescribed
Seizure of stocks	Inspector records action and submits records
Prosecution	Inspector institutes prosecution for breaches
Detection of illegal sale	Inspector conducts necessary enquiries
Inspection records	Copies are submitted to Controlling Authority

2. At manufacturing premises

Inspector's activity	Controlling Authority connection
Inspection of manufacturing premises	Done subject to instructions of Controlling Authority
Examination of manufacturing process	Supervised within regulatory framework

Examination of testing facilities	Inspector reports findings
Sampling	Samples are taken as part of inspection
Inspection report	Detailed report sent to Controlling Authority
Compliance with Act and Rules	Non-compliance is reported

Licensing Authority

Authority appointed by the Central Government to Grant/renew licences and exercising other regulatory powers assigned under the Rules.

Functions of Licensing Authority

Function	Explanation
Grant of licence	Grants licences when prescribed requirements are fulfilled
Renewal of licence	Renews licences after verification of compliance
Verification	Verifies statements/information submitted by applicant
Inspection	Causes manufacturing/testing premises to be inspected where required
Examination of compliance	Determines whether requirements of the Act and Rules are fulfilled
Approval	Grants specified approvals/permissions
Suspension	May suspend licences/permissions in specified circumstances
Cancellation	May cancel licences/permissions in accordance with the Rules
Sampling/testing	Can require samples or testing information in specified cases
Regulatory supervision	Ensures compliance with prescribed standards and conditions

CHAPTER III: IMPORT OF DRUGS AND COSMETICS

The basic objective of the import provisions is to ensure that drugs and cosmetics entering India:

- Conform to prescribed standards of quality;
- Are properly labelled;
- Are not adulterated or spurious;
- Are not harmful or unsafe;
- Are imported only through the prescribed regulatory mechanism;
- Are imported only under the required licence wherever applicable; and
- Do not pose an unacceptable risk to human beings or animals.

Thus, the Act establishes a **control system at the point of entry into India**, with the Customs authorities and drug regulatory authorities working together.

The import provisions are primarily contained in:

Section	Subject
Section 8	Standards of quality
Section 9	Misbranded drugs
Section 9A	Adulterated drugs
Section 9B	Spurious drugs
Section 9C	Misbranded cosmetics
Section 9D	Spurious cosmetics
Section 10	Prohibition of import of certain drugs/cosmetics
Section 10A	Power of Central Government to prohibit import in public interest
Section 11	Customs provisions and powers of Customs officers
Section 12	Power of Central Government to make rules
Section 13	Offences
Section 14	Confiscation
Section 15	Jurisdiction

These provisions together establish the regulatory framework for importation.

CHAPTER III: IMPORT OF DRUGS AND COSMETICS

Standard Quality

According to Section 8, for the purposes of Chapter III:

In case of drugs, a drug is of **standard quality** when it complies with the standards specified in the **Second Schedule**.

In case of cosmetics, a cosmetic is of standard quality when it complies with the standards prescribed under the Rules.

Classes of Drugs/Cosmetics Prohibited from Import

Section 10 is the central provision dealing with **prohibited imports**.

No person shall import the following categories:

A. Drugs or cosmetics not of standard quality.

B. Misbranded drugs: A drug is considered misbranded when, among other things:

- it is coloured, coated, powdered or polished in a way that conceals damage;
- it is made to appear to have greater therapeutic value than it actually has;
- it is not labelled in the prescribed manner; or
- its label, container or accompanying material makes a false or misleading claim.

C. Adulterated drug

A drug is adulterated if,

- it contains filthy, putrid or decomposed material;
- it has been prepared, packed or stored under insanitary conditions;
- its container contains poisonous or deleterious material that could make the drug harmful;
- it contains an unprescribed colouring substance;
- it contains harmful or toxic material; or
- another substance has been mixed with it to reduce its quality or strength.

D. Spurious drug

A drug is spurious if,

- it is imported under the name of another drug;
- it imitates or substitutes another drug in a manner likely to deceive;
- the stated manufacturer is fictitious or non-existent;
- it has been substituted wholly or partly by another drug/substance; or
- it falsely purports to be the product of a particular manufacturer.

CHAPTER III: IMPORT OF DRUGS AND COSMETICS

E. Misbranded or spurious cosmetics

A cosmetic is misbranded if:

- it contains an unprescribed colour;
- it is not labelled as prescribed; or
- its label/container carries a false or misleading statement.

A cosmetic is spurious when,

- it is imported under the name of another cosmetic;
- it imitates or substitutes another cosmetic in a deceptive manner;
- it bears the name of a fictitious manufacturer; or
- it purports to be the product of a manufacturer when it is not actually that manufacturer's product.

F. Section 10 also prohibits import of any drug or cosmetic **for which a licence is prescribed unless it is imported under and in accordance with that licence**.

G. Patent or Proprietary Medicines

A patent or proprietary medicine cannot be imported unless the label/container displays, in the prescribed manner:

- the true formula, or
- the list of active ingredients,
- together with the quantities of those active ingredients.

The purpose is to ensure **transparency regarding composition** and prevent importation of products whose active ingredients are concealed.

H. Drugs Making Prohibited Therapeutic Claims

The Act prohibits import of a drug which, through:

- a statement,
- design,
- device,
- label,
- accompanying material, or
- any other means,

claims to cure, mitigate or prevent diseases or ailments or to have other effects that are prohibited by the Rules.

The Central Government has power under Section 12 to prescribe the diseases or ailments for which imported drugs may not make such claims.

CHAPTER III: IMPORT OF DRUGS AND COSMETICS

I. Unsafe Cosmetics

Import is prohibited when a cosmetic contains an ingredient that may make it **unsafe or harmful when used according to the directions or recommendations**.

S. No.	Category of Cosmetic	Relevant Provision	Import Status / Requirement
1	Cosmetics containing Hexachlorophene	**Rule 134-A**	**Import prohibited**
2	Cosmetics containing lead or arsenic compounds used for colouring purposes	**Rule 135**	**Import prohibited**
3	Cosmetics containing mercury compounds	**Rule 135-A**	**Import prohibited**
4	Cosmetics tested on animals after commencement of the Drugs and Cosmetics (Fifth Amendment) Rules, 2014	**Rule 135-B**	**Import prohibited**

Prohibition by the Central Government in Public Interest – Section 10A

The Government may prohibit import of a drug or cosmetic if it is satisfied that:

1. its use is likely to involve risk to human beings or animals;
2. a drug does not possess the therapeutic value claimed for it; or
3. it contains ingredients in quantities for which there is no therapeutic justification;

and the Government considers prohibition necessary or expedient in the **public interest**.

Such prohibition is made by notification in the **Official Gazette**.

Import of Drugs Under Licence

Category	Import licence
Drugs other than Schedule X	**Form 10**
Schedule X drugs	**Form 10-A**

CHAPTER III: IMPORT OF DRUGS AND COSMETICS

Application for Import Licence

***Drugs other than Schedule X,** application is made in **Form 8**.

***Schedule X drugs,** application is made in **Form 8-A**.

The application is made by:

- the manufacturer having a valid wholesale licence, or
- the manufacturer's agent in India having the appropriate manufacturing or wholesale licence.

*The application is accompanied by the prescribed fee and an undertaking in **Form 9** from/on behalf of the manufacturer.

A copy of the **Registration Certificate in Form 41** is also required with the application.

Registration Certificate – Form 41

The Rules provide for registration of the foreign manufacturer's premises and the drugs intended for import into India.

A Registration Certificate is issued in **Form 41**.

A single Registration Certificate may cover more than one drug or class of drugs manufactured by the same manufacturer when the specified conditions relating to the manufacturing unit are fulfilled.

The purpose of registration is to enable regulatory authorities to identify and assess:

- the foreign manufacturer;
- the manufacturing premises;
- the drugs intended for import; and
- the regulatory status of the products.

Conditions Before Grant of Import Licence

Under **Rule 25A**, before granting Form 10 or Form 10-A, the licensing authority considers whether:

1. the premises where imported substances will be stocked have **proper storage facilities** to preserve the properties of the drugs; and
2. the occupation, trade or business ordinarily carried out by the applicant is appropriate.

The authority may refuse Form 10-A where the applicant has failed to comply with the Act/Rules or is considered unfit.

CHAPTER III: IMPORT OF DRUGS AND COSMETICS

Conditions Attached to an Import Licence

The important conditions include:

1. Compliance with undertaking: The manufacturer must comply with the undertaking furnished in **Form 9**.

2. Inspection: The licensee must allow an authorised Inspector to enter premises where imported substances are stocked, inspect them and take samples.

3. Samples: The licensee must furnish samples from batches when required by the licensing authority.

4. Testing protocols: Where required, the licensee must provide complete protocols of tests applied to the drug.

5. Restriction on sale: If directed by the licensing authority, the licensee cannot sell a batch for which samples/protocols have been requested until the necessary certificate authorising sale is issued.

6. Withdrawal/recall: If a batch fails to conform to prescribed standards of strength, quality or purity, the licensee may be directed to:

- withdraw the remaining batch from sale; and
- recall quantities already distributed, as far as practicable.

7. Records: The licensee must maintain appropriate records of sales of substances for which an import licence is required.

Import of Small Quantities for Examination, Test or Analysis

A special mechanism exists for importing small quantities of drugs that would otherwise be prohibited.

The Rules define a **licence for examination, test or analysis** as a licence in **Form 11** for importing small quantities of otherwise prohibited drugs for examination, test or analysis.

The application is made in **Form 12**. The Rules require the application to be made or countersigned by the head of the institution where the examination, test or analysis will be carried out, or by the proprietor/director of the concerned company or firm.

Important distinction

Form 11 = Licence

Form 12 = Application for that licence

CHAPTER III: IMPORT OF DRUGS AND COSMETICS

Import of Small Quantities for Personal Use

Small quantities of drugs may be imported for personal use under prescribed conditions.

- **Form 12A** – application for permit to import small quantities for personal use;
- **Form 12B** – permit for such import.

The application requires a prescription from a **registered medical practitioner** regarding the need for the drug.

The permit is subject to the conditions prescribed under the Rules and, unless suspended/revoked earlier, is valid for the period specified in the Rules.

Import of New Drugs by Government Hospitals

The Rules also provide a special mechanism for certain new drugs required by Government Hospitals or Autonomous Medical Institutions.

Under **Rule 33A**, small quantities of new drugs whose import would otherwise be prohibited may be imported for treatment of patients suffering from:

- life-threatening diseases;
- diseases causing serious permanent disability; or
- diseases requiring therapies for unmet medical needs.

Important conditions

The drug:

- must be imported under a **licence in Form 11-A**;
- must have been approved for marketing in the country of origin;
- must be used exclusively for the specified patients;
- must be used under medical supervision;
- is subject to inspection and record requirements;
- must be stored appropriately;
- must be dispensed under supervision of a registered pharmacist; and
- normally cannot exceed **100 average doses per patient**, unless the licensing authority permits a larger quantity in exceptional circumstances.

CHAPTER III: IMPORT OF DRUGS AND COSMETICS

Specified Points of Entry for Import of Drugs – Rule 43A

Mode of Import	Specified Point/Area of Entry	Purpose / Route
Rail – Pakistan frontier	Ferozepore Cantonment Railway Station and Amritsar Railway Station	Drugs imported by rail across the frontier with Pakistan
Rail – Bangladesh frontier	Ranaghat, Bongaon and Mohiassan Railway Stations	Drugs imported by rail across the frontier with Bangladesh
Road – Bangladesh	Petrapole Road (West Bengal), Sutarkandi (Assam), Old Raghna Bazar and Agartala (Tripura)	Drugs imported by road from Bangladesh
Road & Rail – Nepal	Raxaul	Drugs imported by road and railway lines connecting Raxaul in India with Birganj in Nepal
Sea	Chennai, Kolkata, Mumbai, Cochin, Nhava Sheva, Kandla, Inland Container Depots at Tuglakabad and Patparganj (Delhi), Tuticorin, Marmugao Port (Goa), and Visakhapatnam	Drugs imported by sea into India
Air	Chennai, Kolkata, Mumbai, Delhi, Ahmedabad, Hyderabad, Goa, Bengaluru and Visakhapatnam	Drugs imported by air into India

Customs Control Over Imported Drugs

Section 11 connects the Drugs and Cosmetics Act with Customs enforcement.

Customs officers have powers regarding drugs and cosmetics whose import is prohibited.

The Customs authorities may:

- detain imported packages suspected of containing prohibited drugs/cosmetics;
- report the detention to the Drugs Controller, India; and
- where necessary, forward the package or sample to the **Central Drugs Laboratory**.

CHAPTER III: IMPORT OF DRUGS AND COSMETICS

Offences and Penalties

Offence	Relevant provision	Punishment
Import of adulterated drug	Section 13(1)(a)	Up to **3 years imprisonment** or **₹5,000 fine** or both
Import of spurious drug	Section 13(1)(a)	Up to **3 years imprisonment** or **₹5,000 fine** or both
Import of spurious/unsafe prohibited cosmetic covered by the provision	Section 13(1)(a)	Up to **3 years imprisonment** or **₹5,000 fine** or both
Other prohibited import under Section 10/Rules	Section 13(1)(b)	Up to **6 months imprisonment** or **₹500 fine** or both
Import contrary to Section 10A notification	Section 13(1)(c)	Up to **3 years imprisonment** or **₹5,000 fine** or both
Subsequent offence under 13(1)(a)/(c)	Section 13	Up to **5 years imprisonment** or **₹10,000 fine** or both
Subsequent offence under 13(1)(b)	Section 13	Up to **1 year imprisonment** or **₹1,000 fine** or both
Offending consignment	Section 14	**Liable to confiscation**
Trial of offence	Section 15	Not below **Metropolitan Magistrate/JMFC**

Confiscation of imported drugs and cosmetics – Section 14

In addition to imprisonment or fine, the **consignment itself can be confiscated**.

Where an offence punishable under Section 13 has been committed, the drug or cosmetic involved in the offence is liable to **confiscation** under Section 14.

CHAPTER IV: MANUFACTURE. SALE AND DISTRIBUTSION OF DRUGS AND COSMETICS

The Important Provisions for Manufacture of Drugs Include:

Provision	Subject
Section 16	Standards of quality
Sections 17, 17A, 17B	Misbranded, adulterated and spurious drugs
Section 18	Prohibition of manufacture and sale of certain drugs and cosmetics
Section 18A	Disclosure of manufacturer
Section 18B	Maintenance of records and furnishing of information
Section 26A	Power of Central Government to prohibit manufacture, etc., in public interest
Section 27	Penalty for manufacture, sale, etc., in contravention of Chapter IV
Section 30	Penalty for subsequent offences
Section 31	Confiscation
Rules 71 onwards	Licensing requirements and conditions for manufacture
Rule 89 onwards	Manufacture for examination, test or analysis
Rules 122A–122E	New drugs and their approval

PART I: PROHIBITION OF MANUFACTURE AND SALE OF CERTAIN DRUGS

Section 18 provides that, from the date specified by the State Government through notification, no person shall himself or through another person:
manufacture for sale or distribution; sell; stock; exhibit; offer for sale; or distribute certain prohibited categories of drugs and cosmetics as mentioned below.

- ✓ **Drugs not of standard quality**
- ✓ **Misbranded drugs**
- ✓ **Adulterated drugs**
- ✓ **Spurious drugs**
- ✓ **Patent or Proprietary Medicines**
- ✓ **Drugs Making Prohibited Therapeutic Claims**
- ✓ **Unsafe Cosmetics**

CHAPTER IV: MANUFACTURE. SALE AND DISTRIBUTSION OF DRUGS AND COSMETICS

PART II: CONDITIONS FOR GRANT OF LICENCE FOR MANUFACTURE

For conventional drugs, the Rules provide requirements relating to:

- Competent technical staff;
- Manufacturing premises;
- Plant and equipment;
- Testing facilities;
- Storage;
- Quality control;
- Good Manufacturing Practices;
- Records and documentation; and
- Inspection by regulatory authorities.

1. Competent Technical Staff – Qualifications

For the relevant **Form 25/Form 25F** licensing framework, the Rules specify qualifications including:

S. No.	Qualification of Technical Staff	Experience Required	Remarks
1	Graduate in **Pharmacy or Pharmaceutical Chemistry**	**18 months' practical experience** after graduation in manufacture of drugs	The required experience may be reduced where the person has undergone prescribed practical training in drug manufacture during the university course.
2	Graduate in **Science with Chemistry as a principal subject**	**3 years' practical experience** after graduation in manufacture of drugs	Experience should be in the manufacture of drugs.
3	Graduate in **Chemical Engineering or Chemical Technology**	**3 years' practical experience** after graduation in manufacture of drugs	Must have appropriate training/experience relevant to drug manufacture.
4	Graduate in **Medicine**	**3 years' practical experience** after graduation in manufacture of drugs	Experience should be relevant to pharmaceutical/drug manufacturing operations.
5	Person possessing a **foreign qualification of comparable standard**	As prescribed/accepted under the Rules	The qualification and training must be considered comparable to the prescribed Indian qualification.

CHAPTER IV: MANUFACTURE. SALE AND DISTRIBUTSION OF DRUGS AND COSMETICS

NOTE: Special Provision for Certain Products

The Rules also contain special provisions for products such as:

- disinfectant fluids;
- insecticides;
- liquid paraffin;
- medicinal gases;
- non-chemical contraceptives;
- plaster of Paris; and
- surgical dressings.

For such products, where pharmaceutical chemistry or pharmacy knowledge is not considered essential, the Licensing Authority may permit manufacture under competent technical staff having adequate experience in manufacturing the particular substance.

2. Manufacturing Premises

The factory premises must comply with the requirements specified in **Schedule M**.

- ✓ Adequate manufacturing space;
- ✓ Suitable plant;
- ✓ Appropriate equipment;
- ✓ Suitable facilities for manufacturing operations; and
- ✓ Facilities appropriate to the type of drug being manufactured.

3. Testing Facilities

The manufacturer must provide and maintain:

- adequate staff;
- appropriate premises;
- laboratory equipment; and
- facilities for testing the **strength, quality and purity** of drugs.

The testing unit is required to be **separate from the manufacturing unit**, and the head of the testing unit is required to be independent of the head of manufacturing.

4. Storage Facilities

The applicant must ensure that the quality of the drug is maintained throughout the period before distribution.

CHAPTER IV: MANUFACTURE. SALE AND DISTRIBUTSION OF DRUGS AND COSMETICS

5. Testing of Patent and Proprietary Medicines

Where a manufacturer applies for a licence to manufacture patent or proprietary medicines, evidence and data must be furnished to justify that the formulation:

1. contains ingredients in appropriate therapeutic/prophylactic quantities;
2. is safe considering the vehicles, excipients, additives and pharmaceutical aids;
3. remains stable under recommended storage conditions; and
4. contains ingredients and quantities for which there is therapeutic justification.

This is particularly important because proprietary formulations may not necessarily be described by a pharmacopeial monograph.

6. Good Manufacturing Practices – Schedule M

Schedule M addresses various aspects of pharmaceutical manufacturing, including:

- premises;
- personnel;
- hygiene;
- production;
- quality control;
- documentation;
- sanitation;
- equipment;
- storage; and
- prevention of contamination and mix-ups.

Thus, obtaining the licence is not the end of regulatory responsibility, the manufacturer must continue to comply with GMP requirements.

CHAPTER IV: MANUFACTURE. SALE AND DISTRIBUTSION OF DRUGS AND COSMETICS

PART III: FORMS OF MANUFACTURING LICENCES

Category	Licence Form
Drugs specified in Schedules C and C(1), excluding specified categories in Part X-B and Schedule X	**Form 28**
Drugs specified in Schedules C, C(1) and X, as applicable under the Rule	**Form 28B**
Large Volume Parenterals, Sera, Vaccines and r-DNA-derived drugs covered by the relevant provision	**Form 28D**
Drugs other than those covered by the Schedule C/C(1) licensing provisions	Appropriate licence under the applicable Rules/Form 25 framework
Loan licence	Appropriate **loan licence form**, e.g. Form 25A/other applicable form

PART IV: CONDITIONS OF LICENCE FOR MANUFACTURE

Once the licence is granted, the manufacturer has to continuously comply with its conditions.

General Conditions of Manufacturing Licence

S. No.	Condition	Requirement / Explanation
1	Maintenance of staff	The licensee must maintain the required **competent technical staff** as prescribed under the Rules.
2	Maintenance of premises	The approved manufacturing premises must be maintained in accordance with the prescribed requirements, including applicable **Schedule M/GMP requirements**.
3	Maintenance of equipment	All necessary **plant, equipment and machinery** required for manufacturing must be properly maintained and kept in working condition.
4	Compliance with the Act and Rules	The licensee must comply with the provisions of the **Drugs and Cosmetics Act, 1940** and **Drugs and Cosmetics Rules, 1945**, along with the conditions of the licence.

CHAPTER IV: MANUFACTURE. SALE AND DISTRIBUTSION OF DRUGS AND COSMETICS

General Conditions of Manufacturing Licence

5	Testing of raw materials	Raw materials used in manufacturing must be appropriately tested for **identity, strength, quality and purity**, as applicable.
6	Testing of finished products	Each batch of the finished drug must undergo the prescribed **quality testing** before release.
7	Maintenance of testing records	Records of tests performed on raw materials and finished products must be maintained in the prescribed manner and for the specified period.
8	Manufacturing records	Complete records relating to the **manufacture of each batch** must be maintained as prescribed, including relevant production details.
9	Compliance with Schedule U	Manufacturing and testing records must be maintained in accordance with the requirements of **Schedule U**, wherever applicable.
10	Inspection by Drug Inspector	The licensee must allow an authorised **Drug Inspector** to inspect the premises, manufacturing operations, testing facilities and relevant records.
11	Sampling	The Inspector may take samples of drugs, raw materials or other relevant materials for **examination, testing or analysis**, as permitted under the Act and Rules.
12	Production of records/information	The licensee must furnish the required **records, documents and information** to the regulatory authorities when requested.
13	Quality control	Adequate arrangements must be maintained to ensure that drugs manufactured are of the required **standard of quality, safety and purity**.
14	Storage	Drugs and materials must be stored under appropriate conditions so that their **quality and stability are maintained**.
15	GMP compliance	The manufacturer must continuously comply with the applicable **Good Manufacturing Practices (GMP)** requirements under Schedule M.

CHAPTER IV: MANUFACTURE. SALE AND DISTRIBUTSION OF DRUGS AND COSMETICS

Duration of Manufacturing Licence

For the **Form 28/Form 28B/Form 28D,** an original or renewed licence is, unless suspended or cancelled earlier, **valid for five years** from the date of grant or renewal.

PART V: MANUFACTURE OF DRUGS FOR TEST, EXAMINATION AND ANALYSIS

The Act recognizes that small quantities of drugs may need to be manufactured for purposes other than commercial sale.

Examples include:

- ✓ laboratory examination;
- ✓ analytical testing;
- ✓ research;
- ✓ formulation evaluation;
- ✓ quality testing;
- ✓ development work; and
- ✓ scientific investigation.

Section 18 specifically provides an exception for manufacture of small quantities for examination, test or analysis, subject to prescribed conditions.

Licence for Manufacture for Test/Examination/Analysis

- **Form 30** is the application for a licence to manufacture drugs for purposes of examination, test or analysis.
- The corresponding licence is issued in **Form 29**. The licence authorizes manufacture of the specified drugs for examination, test or analysis.
- The Form 29 licence is valid for **one year** from the date specified in the licence.

Conditions of Form 29 Licence

Condition	Explanation
Use restricted to examination, test or analysis	Drugs manufactured under Form 29 must be used **exclusively for examination, test or analysis**. They cannot be treated as drugs manufactured for ordinary commercial sale or distribution.
Manufacture at the approved location	The manufacture and the examination, test or analysis must be carried out **at the place specified in the licence**. The licensee cannot arbitrarily shift these activities to an unspecified location.

CHAPTER IV: MANUFACTURE. SALE AND DISTRIBUTSION OF DRUGS AND COSMETICS

Permission for inspection	The licensee must allow an **Inspector appointed under the Act** to enter the premises. The Inspector may enter **with or without prior notice**.
Verification by Inspector	During inspection, the Inspector may verify that the premises are being used **only for examination, test or analysis work** permitted under the licence.
Maintenance of quantity records	The licensee must maintain records showing the **quantity of drugs manufactured** under the licence for examination, test or analysis.
Record of persons receiving the drug	The licensee must maintain details of **any person or persons to whom the manufactured drugs have been supplied**. This provides traceability of the test/analysis material.
Compliance with subsequent requirements	If additional requirements applicable to Form 29 licensees are subsequently prescribed under the Act/Rules, the licensee must comply with them, provided the Licensing Authority gives **at least one month's notice**.
Compliance with Act and Rules	The licence is subject to compliance with the applicable provisions of the **Drugs and Cosmetics Act, 1940 and Drugs and Cosmetics Rules, 1945**. Failure to comply with the licence conditions or applicable provisions can result in regulatory action, including suspension/cancellation.

PART VI: MANUFACTURE OF NEW DRUGS

According to **Rule 122-E**, a **"new drug"** includes a drug that falls into specified categories, principally:

- A drug that has **not been used to a significant extent in India**, and whose use as recommended is **not recognized as safe and effective** by the Licensing Authority.
- A **new dosage form or new route of administration** of a drug already approved.
- A **new indication, new dosage or new combination** of an approved drug, where it falls within the criteria specified under the Rule.
- Certain **modified or new formulations** of an already approved drug, as specified under the Rules.
- A **new fixed-dose combination** of previously approved drugs, subject to the conditions specified in the Rules.

Form 44 is the application for permission to:

- ✓ import a new drug;
- ✓ manufacture a new drug;
- ✓ undertake clinical trials;
- ✓ obtain permission for a fixed-dose combination; or
- ✓ obtain subsequent permission for an already approved new drug.

The Form 44 requires detailed information about the proposed drug.

CHAPTER IV: MANUFACTURE. SALE AND DISTRIBUTSION OF DRUGS AND COSMETICS

Information Required for New Drug Application as per Form 44:

A. Basic drug information

- name of the drug;
- dosage form;
- composition;
- test specifications;
- active ingredients;
- inactive ingredients;
- pharmacological classification;
- proposed indications;
- manufacturer of the bulk drug substance; and
- patent status.

B. Chemical and pharmaceutical information

- drug substance;
- formulation;
- specifications;
- manufacturing;
- quality control.

C. Animal pharmacology & toxicology

- pharmacological effects.
- acute toxicity;
- subacute/chronic toxicity as applicable;
- reproductive toxicity;
- local toxicity;
- carcinogenicity where applicable.

D. Clinical Documentation

The new-drug application includes clinical information covering:

- Phase I human/clinical pharmacology;
- Phase II exploratory clinical trials;
- Phase III confirmatory clinical trials;
- bioavailability;
- dissolution;
- stability;
- regulatory status in other countries; and
- proposed product information.

CHAPTER IV: MANUFACTURE. SALE AND DISTRIBUTSION OF DRUGS AND COSMETICS

Permission/Approval for Manufacture of New Drug Formulation

The Rules provide **Form 46** for permission/approval for manufacture of a new drug formulation under the relevant new-drug provisions.

The form records:

- name of formulation;
- dosage form;
- composition;
- indications; and
- approval/permission details.

Conditions Attached to Approval for New Drug Formulation

Condition	Requirement / Explanation
Compliance with approved specifications	The new drug formulation must conform to the **specifications approved by the Licensing Authority**.
Proper name of the drug	The **proper/generic name** of the drug must be displayed appropriately on the label and must be more conspicuous than the trade/brand name.
Prescription-drug identification	Where applicable, the label must carry the prescribed **red vertical line** indicating that the drug is a prescription medicine.
Prescription warning	The package must carry the prescribed warning indicating that the drug is to be **sold by retail only on the prescription of the specified medical practitioner**.
Post-marketing surveillance	The manufacturer is required to undertake **post-marketing surveillance** during the initial **two years of marketing** of the new drug formulation, in accordance with the approved protocol and requirements.
Reporting of adverse drug reactions	Adverse drug reactions reported during post-marketing surveillance must be communicated to the **Drugs Controller, India and the Licensing Authority**, and any regulatory action arising from such reports must be complied with.
Restrictions on claims	No claim regarding the drug, other than the claims **approved by the Licensing Authority**, may be made without obtaining prior approval.
Approval of packaging materials	Specimens of the **carton, label and package insert** must be approved by the Licensing Authority before the product is marketed.
Compliance with approval conditions	The manufacturer must comply with all other conditions specified in the approval and with the applicable provisions of the **Drugs and Cosmetics Act and Rules**.

LOAN LICENSE

A **loan licence** is a licence that the Licensing Authority may issue to an applicant who does not have his own manufacturing arrangements but intends to use the manufacturing facilities of an existing licensed manufacturer.

Categories of Loan Licence (validity 5 years from the date of grant/ renew.)

Category of Drug	Relevant Rule	Application Form	Manufacturing Facility Utilized
Drugs other than Schedule C, C(1) and X	Rule 69A	Form 24A	Facility of manufacturer holding **Form 25**
Schedule C and C(1) drugs, excluding specified categories	Rule 75A	Form 27-A	Facility of manufacturer holding **Form 28**
Large Volume Parenterals, Sera & Vaccines, r-DNA-derived drugs	Rule 75A(1A)	Form 27DA	Facility meeting the applicable requirements under the Rules

The application is generally made for up to **ten items for each category of drugs**.

Conditions Before Grant of Loan Licence

Requirement	What it means
Adequate equipment	Sufficient equipment to manufacture the proposed drugs
Adequate staff	Appropriate technical/personnel support
Adequate manufacturing capacity	Sufficient capacity to manufacture the drugs on behalf of the applicant
Adequate testing facilities	Facilities for appropriate testing of the drugs

Duplicate Loan Licence

If the Licensing Authority is satisfied that the loan licence has been:

- defaced;
- damaged;
- lost; or
- otherwise rendered useless,

a **duplicate licence** may be issued on payment of the prescribed fee of **₹1,000** for this purpose.

REPACKING LICENSE

Repacking License

Repacking is treated separately from ordinary manufacture under the Rules.

The Rules provide a specific licence for **repacking of drugs for sale or distribution**.

For drugs other than those specified in **Schedules C and C(1)**, the application for a repacking licence is made in **Form 24B** and the licence itself is issued in **Form 25B**

Conditions for Grant/Renewal of Repacking Licence – Rule 71A

Before a **Form 25B** licence is granted or renewed, the applicant must fulfil certain conditions.

Hygienic repacking: The repacking operation must be carried out under **hygienic conditions**, and under the **supervision of a competent person**.

Compliance with Schedule M: The factory premises must comply with the requirements prescribed under Schedule M.

Testing facilities: The applicant must have adequate arrangements in their own premises for testing **strength, quality** and **purity** of the drugs.

The testing unit must be **separate from the repacking unit**.

Single-active-ingredient formulations: The Rules contain a specific provision that an application for a licence for a drug formulation containing a **single active ingredient** is to be made only in the **proper name**, subject to the proviso specified in the Rules.

Conditions of Form 25B Repacking Licence – Rule 74A

Once the licence has been granted, the licensee must continue to comply with the conditions.

S. No.	Condition	Explanation
1	Competent person supervision	Repacking must always be carried out under the personal supervision of at least one person approved as a competent person by the Licensing Authority.
2	Testing arrangements	The licensee must maintain adequate arrangements in their own premises for testing the strength, quality and purity of repacked drugs, or use an institution approved by the Licensing Authority for such testing.
3	Storage	Adequate arrangements must be made for storage of drugs.
4	Compliance with Act and Rules	The licensee must comply with the Drugs and Cosmetics Act, Rules and subsequent applicable requirements.

REPACKING LICENSE

5	Inspection	The Inspector may enter the premises with or without notice, inspect the premises and take samples of repacked drugs.
6	Testing of raw material	Each batch/lot of raw material used for repacking must be tested.
7	Testing of repacked product	Each batch of the repacked product must also be tested.
8	Test records	Records/registers of testing must be maintained according to Schedule U.
9	Retention of records	Test records must be retained for five years from the date of repacking.
10	Inspection of records	The licensee must allow the Inspector to inspect the records/registers and provide information required to verify compliance.
11	Inspection Book	An Inspection Book in Form 35 must be maintained for recording observations and defects noticed by the Inspector.
12	Reference samples	Reference samples from each batch must be retained in the quantity and for the period prescribed by the Rules.

Reference Samples in Repacking

The licensee must maintain **reference samples from each batch** in a quantity at least **twice the quantity required to conduct all tests performed on that batch**.

Retention period

Drug	Reference Sample Retention
Drug having an expiry date on the label	3 months beyond the expiry date of potency
Drug without expiry date of potency specified	3 years from date of manufacture

Competent Person for Repacking

The Rules also explain who can be considered a **competent person** for purposes including Rule 71A and Rule 74A.

A person holding a **Diploma in Pharmacy approved by the PCI**, or a person registered under the Pharmacy Act, qualifies under one route. Other prescribed routes include specified educational qualifications and practical experience in manufacture, dispensing or repacking of drugs.

SCHEDULE G

Schedule G is specifically linked with **Rule 97**. The Schedule contains a list of specified substances and also covers certain preparations containing those substances.
The regulatory importance of Schedule G is primarily related to **labelling and medical supervision**. The key requirement is that medicines for internal use containing a substance specified in Schedule G must carry a prescribed cautionary statement indicating that the preparation is dangerous to take except under medical supervision.

Rule 97 is titled **"Labelling of medicines"**
For a medicine intended for **internal use**, **Rule 97(1)(a)** states that if it contains a substance specified in Schedule G, its container must bear the prescribed cautionary statement.
"Caution: it is dangerous to take this preparation except under medical supervision"
The document further requires this statement to be:

- **conspicuously printed**, and
- **surrounded by a line**, within which there shall be **no other words**.

- Aminopterin
- L-Asparaginase
- Bleomycin
- Busulphan and its salts
- Carbutamide
- Chlorambucil and its salts
- Chlorothiazide and other derivatives of 1,2,4-benzothiadiazine
- Chlorpropamide and its salts
- Chlorthalidone and other derivatives of the chlorobenzene compound
- Cis-Platin
- Cyclophosphamide and its salts
- Cytarabine
- Daunorubicin
- Di-Isopropyl Fluorophosphate
- Disodium Stilboestrol Diphosphate
- Doxorubicin Hydrochloride
- Ethacrynic Acid and its salts
- Ethosuximide
- Glibenclamide
- Hydantoin, its salts and derivatives and their salts
- Hydroxyurea
- Insulin, all types
- Lomustine Hydrochloride
- Mannomustine and its salts
- Mercaptopurine and its salts
- Troxidone
- Metformin and its salts
- Methsuximide
- Mustine and its salts
- Paramethadione
- Phenacemide
- Phenformin and its salts
- 5-Phenylhydantoin and its alkyl/aryl derivatives and salts
- Primadone
- Procarbazine Hydrochloride
- Quinthazone
- Sarcolysine
- Sodium-2-Mercaptoethanesulfonate
- Tamoxifen Citrate
- Testolactone
- Thiotepa
- Tolbutamide
- Tretamine and its salts

SCHEDULE H

Schedule H contains drugs that are required to be supplied by retail only against the prescription of a Registered Medical Practitioner, subject to the conditions prescribed in the Rules 65 & 97.

Importance: To place additional controls over the **retail supply and labelling of prescription medicines**.

NOTE: The Rules 65; specifically state that substances specified in **Schedule H, Schedule H1 or Schedule X shall not be sold by retail except on and in accordance with the prescription of a Registered Medical Practitioner**.

Supply to Doctors, Hospitals, Dispensaries and Nursing Homes

The Rules also regulate supply of Schedule H, H1 or X drugs to professional/institutional users **against a signed written order**. The licensee must preserve that written order for **two years**.

What Constitutes a Valid Prescription?

The prescription must:

1. Be in writing

It must be **written and signed** by the person giving it using his usual signature.

2. Be dated

The prescription must bear the **date**.

3. Identify the patient

It must specify name and address of the person for whose treatment it is given; or, in veterinary use, name and address of the animal's owner.

4. Specify quantity and dose

The prescription must indicate total amount of medicine to be supplied; and dose to be taken.

Dispensing of Schedule H Prescription

Under Rule 65(11); The prescription:

1. must not be dispensed more than once, unless the prescriber specifically states that it may be dispensed more than once;
2. if the prescription specifies a particular number of times or intervals, it must be dispensed according to those directions;
3. at the time of dispensing, the seller must record on the prescription:
    1. name and address of the seller; and
    2. date on which the prescription was dispensed.

SCHEDULE H

No Substitution by the Dispenser

The Rules state that a person dispensing a prescription containing Schedule H, H1 or X substances **cannot supply another preparation in lieu of the prescribed preparation**, whether or not it contains the same substance.

Record-Keeping for Schedule H

Particular	Record required
Serial number	Yes
Date of supply	Yes
Prescriber's name and address	Yes
Patient's name and address	Yes
Name of drug	Yes
Quantity supplied	Yes
Manufacturer	For Schedule C/H/H1 categories as specified
Batch number	Applicable
Expiry/potency date	Where applicable
Pharmacist's signature	Yes

Labelling of Schedule H Drugs

Under **Rule 97**, a medicine for internal use containing a Schedule H substance must carry the prescribed prescription-drug labelling. The label must carry **symbol "Rx"** and the symbol must be **conspicuously displayed at the left top corner of the label**.

In addition, the label must carry the warning:

"Schedule H drug – Warning: To be sold by retail on the prescription of a Registered Medical Practitioner only."

NOTE: The Rules make a special distinction where a Schedule H substance also falls within the purview of the **Narcotic Drugs and Psychotropic Substances Act, 1985**. In that situation, the label must carry the symbol **"NRx".**

The symbol must be:

- in red;
- conspicuously displayed;
- at the left top corner of the label.

The same Schedule H prescription warning is also required.

Kindly Note

Schedule H	**Rx**
Schedule H + NDPS applicability	**NRx** in red
Schedule H1	**Rx** in red + special warning box
Schedule X	**XRx** in red

Red Vertical Line

There is another important labelling requirement. The Rules require the labels of specified categories, including drugs falling under **Schedules G, H and X**, to carry a **conspicuous red vertical line** on the **left side of the label**, running throughout the body of the label. The line should be **not less than 1 mm in width**.

Advertising of Schedule H Drugs

No advertisement of drugs specified in Schedule H, Schedule H1 or Schedule X shall be made except with the previous sanction of the Central Government.

Schedule H and Manufacturing Licence Conditions

Schedule H also appears in the conditions imposed on manufacturing licences.

The manufacturer must comply with the relevant manufacturing and quality-control requirements, and the Rules specifically restrict advertising of Schedule H/H1/X drugs unless prior Central Government sanction has been obtained.

Schedule H and Inspection

The licensing provisions require licensees to produce registers and records for inspection when required by an Inspector.

The Inspector may require information necessary to determine whether the Act and Rules have been complied with.

Therefore, for Schedule H drugs, documentation is an important part of regulatory compliance.

Examples of Schedule H Drugs

- Abacavir
- Abciximab
- Acamprosate calcium
- Acebutol Hydrochloride
- Aclarubicin
- Albendazole
- Acyclovir
- Adenosine
- Adrenocorticotrophic hormone
- Alendronate sodium
- Allopurinol
- Alprazolam
- Alprostadil
- Amantadine Hydrochloride
- Amikacin sulphate
- Amiloride hydrochloride
- Amitriptyline
- Amlodipine
- Antibiotics
- Artemether
- Artesunate
- Atorvastatin
- Azathioprine
- Aztreonam

SUMMARY

Feature	Schedule H	Schedule H1	Schedule X
Basic classification	Prescription drugs	Drugs requiring additional control	More strictly controlled prescription drugs
Retail sale	Prescription of Registered Medical Practitioner	Prescription required	Prescription required
Label symbol	Rx	Rx in red + special warning box	XRx in red
Special record	Prescription documentation	Separate register with prescriber, patient, drug and quantity	Additional record/control requirements
Prescription retention	General prescription requirements	Separate H1 records maintained for 3 years	Prescription in duplicate; one copy retained for 2 years
Written order for institutions	Required	Required	Required
Advertising	Prior Central Government sanction	Prior Central Government sanction	Prior Central Government sanction
Storage	Ordinary prescription-drug controls	Additional monitoring	Specific secure storage provisions
Relative control	High	Higher monitoring for specified drugs	More stringent

SCHEDULE M: Good Manufacturing Practices and Requirements of Premises, Plant and Equipment for Pharmaceutical Products

It is linked with **Rules 71, 74, 76 and 78**.The basic philosophy is that a pharmaceutical manufacturing facility must have documented systems, suitable premises, appropriate equipment, trained personnel, controlled manufacturing processes and adequate quality-control arrangements so that the finished drug consistently meets the required standards.

Why is Schedule M important?

Schedule M is essentially the **GMP framework for pharmaceutical manufacturing premises.** It controls the entire manufacturing environment:

Objective is to prevent:

- contamination,
- cross-contamination,
- mix-ups,
- deterioration of materials,
- manufacturing errors,
- inadequate testing,
- improper storage,
- and release of defective products.

Location
↓
Building & premises
↓
Personnel
↓
Materials
↓
Equipment
↓
Manufacturing operations
↓
Documentation
↓
Quality assurance
↓
Quality control
↓
Self-inspection/audit

NOTE: The conditions for grant/renewal of manufacturing licences require the factory premises to comply with Schedule M. The applicant must also provide adequate space, plant and equipment for manufacturing operations, with the relevant requirements specified in Schedule M.

Overall Structure of Schedule M

Major Area	What it Controls
General requirements	Location, building, premises and manufacturing environment
Plant and equipment	Design, installation, movement, maintenance and validation
Personnel	Qualification, hygiene, health and training
Production/manufacturing controls	Material handling, processing and prevention of mix-ups
Storage/warehousing	Raw materials, packaging materials and finished products
Documentation & records	Traceability and manufacturing history
Labels & printed materials	Prevention of labelling and packaging errors
Quality assurance	Overall system ensuring product quality
Self-inspection & quality audit	Internal evaluation and corrective action
Quality control	Testing, release/rejection and specifications
Specific manufacturing requirements	Special dosage forms/products
Sterile products	Additional requirements for aseptic/sterile manufacturing

PART I: GOOD MANUFACTURING PRACTICES FOR PREMISES AND MATERIALS

Location and surroundings: The factory should be located and designed in a way that minimizes the risk of contamination from the external environment.

The document specifically identifies possible external sources such as:

- open sewage;
- drains;
- public lavatories;
- factories producing obnoxious odours/fumes;
- excessive soot;
- dust;
- smoke;
- chemical emissions; and
- biological emissions.

Buildings and Premises

The factory building must be:

- appropriately designed;
- properly constructed;
- adapted for the manufacturing operation;
- adequately maintained; and
- capable of permitting production under hygienic conditions.

The building must also comply with applicable requirements of the **Factories Act, 1948**.

Adequate Working Space

Manufacturing, processing, warehousing, packaging, labelling and testing areas must have sufficient working space.

The arrangement should permit:

- orderly placement of equipment;
- logical movement of materials;
- proper movement of personnel;
- prevention of mix-ups;
- prevention of contamination; and
- prevention of cross-contamination.

Pest Control

The premises should be designed and maintained to prevent entry of:

- insects;
- pests;
- birds;
- vermin; and
- rodents.

Walls, floors and ceilings should be:

- smooth;
- free from cracks; and
- capable of being easily cleaned, painted and disinfected.

Lighting, Ventilation and Air Control

Production and dispensing areas should be:

- adequately illuminated;
- effectively ventilated;
- provided with appropriate air-control facilities;
- air-conditioned where required by the operation/dosage form;
- provided with Air Handling Units where applicable.

Temperature and, where necessary, humidity must be maintained according to the requirements of the product.

The relevant environmental conditions must also be **regularly monitored**.

Drainage

Drainage systems must be:

- adequately sized;
- appropriately designed;
- suitable for the category of product; and
- designed to prevent backflow.

They should also prevent insects and rodents from entering the premises.

Plant and Equipment

Schedule M requires the design of the plant to be appropriate to:

- the nature of the manufacturing activity; and
- the quantum/scale of activity.

Equipment should be installed so that:

- materials can flow logically;
- unnecessary crossing of personnel is avoided;
- manufacturing operations can be performed efficiently;
- cleaning is possible; and
- contamination risks are minimized.

Restricted Access

Entry into manufacturing sections should be controlled.

People who are **not associated with the relevant manufacturing activity should not have unrestricted access** to manufacturing areas.

This is particularly important for:

- sterile areas;
- potent drug areas;
- hazardous drug areas;
- controlled environments; and
- areas where cross-contamination is a concern.

Personnel Hygiene

Schedule M places considerable importance on personal hygiene.

Facilities should be available for:

- personal cleanliness;
- changing clothes;
- changing footwear;
- toilets;
- appropriate protective clothing;
- head covers;
- gloves.

Separate arrangements for men and women workers are specified for toilets and changing from outside clothing/footwear to factory clothing/footwear.

Prohibited Activities in Manufacturing Areas

The manufacturing premises should not be used for:

- dining;
- smoking;
- spitting;
- chewing;
- littering;
- inappropriate food storage.

Separate areas should be available for personnel to take food or rest.

Personnel with Infectious Disease

A person suffering from a **contagious disease** should not be involved in manufacturing activities.

This protects both:

- the product; and
- other personnel.

Standard Operating Procedures (SOPs)

An SOP ensures that an operation is not dependent solely on an individual's memory or personal method. Written SOPs should cover activities associated with drug manufacture, including:

- cleaning;
- sanitation;
- personal hygiene;
- plant maintenance;
- equipment maintenance;
- procurement;
- quarantine;
- testing;
- warehousing;
- handling of materials.

Equipment Validation

The performance of equipment and systems must be appropriately **validated**, and their use must be monitored.

This means the manufacturer must have confidence that equipment performs consistently and produces reliable results.

Examples:

- weighing balance → calibration;
- tablet machine → performance verification;
- autoclave → sterilization-cycle validation;
- HVAC → qualification/performance monitoring.

Separate Areas for Ancillary Activities

Schedule M requires separate dedicated areas for activities such as:

- receipt of raw materials;
- cleaning;
- warehousing;
- issue of raw materials;
- packaging materials;
- containers and closures;
- finished products.

Identification and Status Labelling of Materials

Materials should have proper:

- identification;
- control numbers;
- inventory tags; and
- status labels.

The status of material should be clear—for example:

- ❖ **QUARANTINE**
- ❖ **APPROVED**
- ❖ **REJECTED**

This helps prevent accidental use of unapproved materials.

Prevention of Mix-ups

Schedule M emphasizes prevention of mix-ups at every stage. There should be SOPs and appropriate arrangements for preventing:

- wrong material;
- wrong batch;
- wrong label;
- wrong packaging material;
- wrong product;
- wrong status.

Materials with strong odours should be stored in tightly closed containers and appropriately segregated.

Documentation and Records

This is one of the **most important components of Schedule M**. Documentation is an essential part of the quality-assurance system.

Its objectives include:

1. defining specifications;
2. defining manufacturing methods;
3. defining control procedures;
4. ensuring personnel have necessary information;
5. supporting batch-release decisions; and
6. providing an **audit trail** for investigation of defective batches.

Principles of Documentation

Documents should be:

- prepared appropriately;
- reviewed;
- approved;
- signed;
- dated;
- clearly identified;
- legible;
- regularly reviewed;
- kept up to date.

Any alteration to an entry should be appropriately **signed and dated**.

Important GMP principle

If an activity is not documented, it becomes difficult to demonstrate that it was performed.

Electronic Records

The Schedule allows electronic data-processing systems, but appropriate controls must be in place.

The document requires, among other things:

- authorized access;
- records of changes/deletions;
- restricted access through passwords or equivalent controls;
- independent checking of critical data;
- backup of electronic batch records;
- ready availability of relevant data during the retention period.

Labels and Printed Materials

Labels are considered essential for:

- identification of drugs; and
- proper use of drugs.

They should be printed clearly and legibly and contain prescribed product information.

Different status labels may be used to distinguish:

***Under test** ***Approved/passed** ***Rejected**

Prevention of Packaging/Label Mix-up

Printed packaging materials and leaflets for different products should be stored separately.

Before release, labels, cartons, boxes, inserts and leaflets should be examined by the **Quality Control Department**.

Before packaging/labeling a batch:

Samples must be drawn from the bulk and appropriately tested, approved and released by Quality Control personnel.

Quality Assurance

Schedule M defines **Quality Assurance** as the totality of arrangements made to ensure that products have the quality required for their intended use.

The QA system should ensure:

- appropriate product design/development;
- correct starting and packaging materials;
- controls on starting/intermediate/bulk products;
- in-process controls;
- calibration and validation;
- correct processing;
- final checking;
- authorized batch release.

Quality Control

The Quality Control system is responsible for ensuring that materials and products meet their required specifications.

The Schedule identifies important QC responsibilities, including:

- evaluating finished products;
- evaluating raw and semi-finished materials when necessary;
- reviewing production records;
- investigating errors;
- approving/rejecting procedures and specifications;
- establishing shelf life;
- reviewing complaints;
- reviewing recalls;
- reviewing returned products;
- reviewing Master Formula Records.

Batch Release

A batch should **not be released for sale/supply until authorized personnel certify that it has been manufactured and controlled according to established requirements.**

So:

Manufacturing completed ≠ Automatically released

There must be appropriate **quality review and authorization**.

Self-Inspection and Quality Audit

Schedule M requires the concept of **self-inspection** to evaluate compliance with GMP.

An independent and experienced team should objectively assess implementation of:

- methodology;
- procedures;
- production practices;
- quality-control practices.

The inspection should be documented and should include:

- observations;
- evaluation;
- conclusions;
- corrective actions;
- follow-up.

Areas Covered During Self-Inspection

- Area
- Personnel
- Premises
- Personnel facilities
- Building/equipment maintenance
- Storage of starting materials
- Storage of finished products
- Equipment
- Production
- In-process control
- Quality control
- Documentation
- Sanitation and hygiene
- Validation/revalidation
- Calibration
- Recall procedures
- Complaint management
- Label control
- Previous self-inspections and corrective actions

Complaints and Product Recall

Schedule M's GMP framework also covers:

- product complaints;
- returned products;
- product recalls;
- investigation of quality problems.

These are important because GMP does not end when the product leaves the factory. The manufacturer must have systems to identify and respond to problems arising after distribution. The quality-control responsibilities specifically include reviewing complaints, recalls and returned/salvaged products.

Master Formula Record

The Master Formula Record is a key manufacturing document.

It provides the approved master instructions for manufacturing and controlling a product.

The uploaded document's specific requirements include information such as:

- product name;
- generic name;
- strength;
- dosage form;
- containers;
- packaging materials;
- labels and closures;
- identity and quantity of raw materials;
- equipment/vessels;
- critical processing instructions;
- theoretical yield;
- permissible yield limits;
- precautions;
- in-process testing requirements;
- responsible personnel/departments.

Cleanroom Grades

The sterile manufacturing section classifies areas according to environmental requirements.

Aseptic preparations

Grade	Typical operation
Grade A	Aseptic preparation and filling
Grade B	Background environment for Grade A activities
Grade C	Preparation of solution to be filtered
Grade D	Handling of components after washing

Environmental Monitoring

Environmental parameters must be established and periodically monitored.

Parameter	Frequency
Air particulate monitoring	**6-monthly**
HEPA filter integrity/smoke testing	**Yearly**
Air-change rate	**6-monthly**
Air-pressure differential	**Daily**
Temperature & humidity	**Daily**
Microbiological monitoring in aseptic areas	**Daily**

Garments in Aseptic Areas

- outdoor clothing should not enter sterile areas;
- garments should be non-shedding;
- tightly woven material should be used;
- cotton garments should not be used;
- garments should protect the product from contamination;
- appropriate head covering and protective clothing are required;
- clean, sterilized protective garments should be used;
- masks and gloves are changed according to the specified operations.

Special Areas for Parenteral Manufacturing

For parenteral products, the document describes separate areas such as:

1. Water management area
2. Container/closure preparation area
3. Solution preparation area
4. Filling/capping/sealing area
5. Sterilization area
6. Quarantine area
7. Visual inspection area
8. Packaging area

Repacking under Schedule M

Schedule M specifically addresses **repacking of drugs and pharmaceutical chemicals**.

The document recommends:

- powder disintegrator;
- electrically operated powder sifter;
- stainless-steel scoops and vessels;
- weighing/measuring equipment;
- filling equipment;
- electric sealing machine.

A minimum basic installation area of **30 m^2** is recommended, with an appropriate exhaust system where fine floating powder particles are generated.

Important Exception in Schedule M

The Schedule acknowledges that certain categories do not have specific space/equipment requirements prescribed within Schedule M.

The document specifically mentions categories such as:

chemicals and pharmaceutical aids;

- gauzes and bandages;
- medicinal gases;
- empty gelatin capsules;
- non-chemical/mechanical contraceptives;
- diagnostic kits and reagents;
- medical devices;
- new dosage forms and delivery systems;
- disinfectant fluids;
- antacids;
- certain veterinary biologicals;
- repacking of drugs, etc.

For such categories, the Licensing Authority may modify/relax or alter requirements where justified by the nature and extent of manufacturing operations, with reasons recorded in writing.

The "5M" Way to Remember Schedule M

1. MAN
Personnel
Health
Hygiene
Training

2. MACHINE
Equipment
Installation
Maintenance
Calibration/validation

3. MATERIAL
Receipt
Quarantine
Testing
Approval/rejection
Storage

4. METHOD
SOPs
Manufacturing process
In-process controls
Packaging

5. MONITORING
Quality Control
Documentation
Validation
Self-inspection
Complaints/recall

Schedule M vs Schedule M-II vs Schedule M-III

Schedule	Deals with
Schedule M	GMP + premises, plant and equipment for pharmaceutical products
Schedule M-II	Requirements of factory premises for manufacture of cosmetics
Schedule M-III	Quality Management System for notified medical devices and in-vitro diagnostics

Schedule M ensures that the right product is manufactured, by the right process, using the right materials and equipment, in the right environment, by trained personnel, with adequate documentation and testing, and is released only after appropriate quality evaluation.

SCHEDULE P: Life Period of Drugs

This schedule is specifically linked with **Rule 96**.
The Schedule specifies:

1. **Name of the drug**
2. **Maximum period in months** between the date of manufacture and date of expiry
3. **Conditions of storage** under which that life period applies.

What does "Life Period" mean?

The Schedule defines the permitted period between:
Date of manufacture → Date of expiry
during which the **labelled potency period of the drug shall not exceed the period prescribed in Schedule P**, provided the drug is stored under the specified conditions.

Basic Structure of Schedule P

Schedule P has four principal columns:

Column	Particular
1	Serial number
2	Name of drug
3	Period in months between manufacture and expiry
4	Condition of storage

Why is Schedule P important?

Schedule P is important because **expiry date is not arbitrary**. The permitted period has to comply with the applicable regulatory requirements.

Drug

↓

Identify applicable Schedule P entry

↓

Determine maximum permitted life period

↓

Follow specified storage condition

↓

Assign expiry date

↓

Print manufacturing + expiry information on label

The labelling provisions specifically connect Schedule P with the expiry period.

Conditions of Storage are Important

Terms Used:

- **Cool place** means place having a temperature between 10℃ to 25℃.
- **Cold place** means place having a temperature less than 8℃.
- In a well closed container, protected from light, in a cool place.
- At a temperature between 2℃ to 8℃ and must not be allowed to freeze.
- Capsules be stored in well closed container below 30℃.
- Store at room temperature, if storage condition are not specified.

Therefore, **Shelf life cannot be considered independently of storage conditions.**

Schedule P and Stability

Schedule P is fundamentally related to **stability of pharmaceutical products**.

A drug can undergo:

- chemical degradation;
- physical changes;
- loss of potency;
- changes in appearance;
- changes in dissolution;
- microbial deterioration.

Therefore, the expiry period should represent the period during which the product is expected to remain within its specified quality requirements under the prescribed storage conditions.

Schedule P and Labelling

Drugs specified in Schedule P and their preparations, including combinations with other drugs, shall bear on their labels the date of manufacture and the date of expiry of potency.

The period between manufacture and expiry **shall not exceed the period laid down in Schedule P**, under the storage conditions specified there.

Therefore:

For Schedule P drugs:

Label must show

1. **Date of manufacture**
2. **Date of expiry of potency**

Schedule P vs Drugs Not Included in Schedule P

Drugs not included in Schedule P must bear:

- date of manufacture; and
- date of expiry,

and the expiry period **shall not exceed 60 months from the date of manufacture**, subject to the provision concerning extension by the Licensing Authority.

Feature	Schedule P drugs	Drugs not included in Schedule P
Date of manufacture	Required	Required
Date of expiry	Required	Required
Maximum period	As specified in Schedule P	**Not more than 60 months**, according to the uploaded text
Storage condition	As specified in Schedule P	Applicable requirements
Extension	Subject to applicable regulatory provision	Licensing Authority may extend if satisfactory evidence is produced

NOTE: Extension of Expiry Period

The period may be extended by the **Licensing Authority** for a specified drug if the manufacturer produces **satisfactory evidence justifying the extension.**

Schedule PI: Pack size of drugs.

Examples from Schedule P

Drug	Life period	Storage condition stated in document
Adramycin	**30 months**	In a cool place
Ampicillin	**36 months**	In a cool place
Ampicillin Capsules	**24 months**	—
Ampicillin Sodium	**36 months**	In a cool place
Ampicillin Trihydrate	**30 months**	In a cool place
Amoxycillin Trihydrate	**36 months**	In a cool place
Amoxycillin Trihydrate Capsules	**24 months**	—
Amoxycillin Trihydrate Dry Syrup	**18 months**	—
Bacitracin	**18 months**	In a cool place
Bacitracin/Zinc Bacitracin Tablets	**12 months**	—
Bacitracin Lozenges	**12 months**	—
Carbenicillin Sodium Injection	**24 months**	Temperature not exceeding 5°C
Carbenicillin Sodium Powder	**24 months**	Temperature not exceeding 5°C
Cephalexin	**24 months**	In a cool place
Chloramphenicol	**60 months**	In a cool place

SCHEDULE U: Particulars to be Shown in Manufacturing Records and Records of Tests

Schedule U prescribes the particulars that must be recorded in manufacturing records and testing records of drugs. The Schedule is expressly referred to under **Rules 74, 74A, 74B, 78 and 78A.**

The records referred to in Rule 74 are required to be retained for **five years from the date of manufacture**.

Why is Schedule U Important?

Schedule U provides traceability of the manufacturing and testing history of a drug batch. It allows the manufacturer and regulatory authorities to answer questions such as:

- What product was manufactured?
- Which batch was it?
- What raw materials were used?
- How much of each ingredient was used?
- Who weighed the ingredients?
- Who checked them?
- When was manufacturing started?
- What environmental conditions existed?
- What in-process tests were performed?
- What was the theoretical yield?
- What was actually produced?

I. PARTICULARS TO BE SHOWN IN MANUFACTURING RECORDS

A. Substances Other Than Parenteral Preparations

1. Serial Number
2. Name of the Product
3. Reference of Master Formula Record
4. Lot/Batch Size
5. Lot/Batch Number
6. Date of commencement of manufacture and date of completion of manufacture and assigned date of expiry.
7. Name of all ingredients, specifications, quantities required, and quantities actually used. All weighing and measurements shall be carried out by a responsible person and shall be counter checked and signed by the competent technical staff under whose personal supervision the ingredients are used for manufacture.
8. Control numbers of raw material used in the formulation.
9. Date, time and duration of mixing.
10. Details of environmental controls like RT, RH.
11. Date of granulation, wherever applicable.
12. Theoretical weight and actual weight of granules/ powder blend.
13. Records of In-process control (Periodically whenever necessary)
    - Uniformity of mixing
    - Moisture content of granules/ powder in case of tablet/ capsules.
    - pH of solution in case of liquid
    - Weight variation
    - Disintegration time
    - Hardness
    - Friability test
    - Leak test in case of strip packing
    - Filled volume of liquids
    - Quantity of tablets/ capsule in the final container.
    - Content of ointment in the filled containers.
14. Date of compression in case of tablets/ date of filling in case of capsules
15. Date of sealing/ coating/ polishing in case of capsules/ tablets wherever applicable.
16. Reference to analytical report number stating the result of test and analysis.
17. Separate records of the disposal of the rejected batches and of batches withdrawn from the market.

18. The theoretical yield and actual production yield and packing particulars indicating the size and quantity of finished packings.
19. Specimen of label/ strip, carton with batch coding information like batch number, date of manufacture, date of expiry, retail price as applicable stamped thereon and inserts used in finished packings.
20. Signature with date of competent technical staff responsible for the manufacture
21. Counter- signature of the head of the testing units or other approved person-in-charge of testing for having verified the batch records and for having released and batch for sale & distribution, the quantity released and date of release.
22. Dare of release of finished packings and quantity released for sale and distribution.
23. Quantity transferred to warehouse.
24. For hypodermic tablets and ophthalmic preparations, which are required to be manufactured under aseptic conditions, records shall be maintained indicating the precautions taken during the process of manufacture to ensure that aseptic conditions are maintained.

II. PARTICULARS TO BE SHOWN IN MANUFACTURING RECORDS OF PARENTERAL PRODUCTS

1. Serial Number
2. Name of the Product
3. Reference of Master Formula Record
4. Lot/Batch Size
5. Lot/Batch Number
6. Date of commencement of manufacture and date of completion of manufacture and assigned date of expiry.
7. Name of all ingredients, specifications, quantities required, and quantities actually used. All weighing and measurements shall be carried out by a responsible person and shall be counter checked and signed by the competent technical staff under whose personal supervision the stocks are issued and by another competent technical staff whose supervision the ingredients are used for manufacture.
8. Control numbers of raw material used in the formulation.
9. Date, time and duration of mixing.
10. Details of environmental controls like RT, RH, microbial count in the sterile working areas.
11. pH of the solution, wherever applicable
12. Date and method of filtration
13. Sterility test, reference on bulk batch wherever applicable.

14. Record of check on volume filled.
15. Date of filling.
16. Records of tests employed:
- To ensure the sealed ampoules are leak proof.
- To check the presence of foreign particles.
- Pyrogen test, wherever applicable
- Toxicity test, wherever applicable

17. Records of checking of instruments and apparatus of sterilization (indicators).
18. Records of cleaning and sterilization of containers and closures, if necessary.
19. Records of sterilization in case of parenteral preparations which are heat sterilized including particulars of time, temperature and pressure employed. Such records should be marked to relate to the batch sterilized.
20. Number and size of containers filled and quantity rejected.
21. The theoretical yield and actual yield and the percentage yield thereof.
22. Reference to analytical report numbers stating whether of standard quality or otherwise.
23. Specimen of labels, cartons, etc, with batch coding information like batch number, date of manufacture, date of expiry, as applicable, stamped thereon, and inserts used in finished packing.
24. Signature with date of competent technical staff responsible for the manufacture
25. Particulars regarding the precautions taken during the manufacture to ensure that aseptic conditions are maintained.
26. Countersignature of head of the testing unit or person in charge of testing for having verified the documents and for having released the product for sale and distribution, the quantity released and date of release.
27. Records for having transferred to warehouse giving packings and quantities.
28. Separate records of the disposal of the rejected batches and of all batches withdrawn from the market.
29. Records of reprocessing if any and particulars of reprocessing.

III. PARTICULARS TO BE SHOWN IN RECORDS OF RAW MATERIALS

1. Date of receipt
2. Invoice number
3. Name & address of the manufacturer/ supplier
4. Batch number
5. Quantity received
6. Pack size
7. Date of manufacture
8. Date of expiry, if any, date of analysis and release/ rejection by quality control.
9. Analytical report number with special remarks, if any.
10. Quantity issued and date of issue
11. The particulars of the name and batch numbers of products for the manufacture of which issued.
12. Particulars of the proper disposal of the stocks.

PARTICULARS TO BE RECORDED IN THE ANALYTICAL RECORDS

1. Tablets and Capsules

- Analytical Report Number
- Name of the Sample
- Date of receipt of sample
- Batch/ Lot Number
- Protocols of tests applied- Records regarding various tests applied (including readings & calculations) should be maintained with necessary reference whenever necessary.
- Signature of the analyst
- Opinion & signature of the approved analyst.

2. Parenteral Preparations

- Analytical report number
- Name of the sample
- Batch number
- Date of receipt of samples
- Number of containers filled
- Number of containers received
- Protocols of test applied: Clarity, pH, identification, volume in container, sterility, pyrogen test, toxicity test, results of assay. Records regarding various tests applied (including readings & calculations) should be maintained with necessary reference whenever necessary.
- Signature of the analyst
- Opinion & signature of the approved analyst.

3. Raw Materials

- Serial Number
- Name of the materials
- Name of the manufacturer/ supplier
- Quantity received

- **Invoice/ challan number and date**
- **Protocols of test applied-** Records regarding various tests applied (including readings & calculations) should be maintained with necessary reference whenever necessary.

4. Container, Packing Material

- Serial Number
- Name of the item
- Name of the Manufacturer/ supplier
- Quantity received
- Invoice/ challan number and date
- Results of test applied- Records regarding various tests applied (including readings & calculations) should be maintained with necessary reference whenever necessary.
- Remarks
- Signature of the examiner

Schedule U – MFR vs BMR

Schedule U specifically requires the manufacturing record to contain the **reference to the Master Formula Record**.

Master Formula Record (MFR)	**Batch Manufacturing Record / Schedule U record**
Master/approved instructions	Actual batch history
Defines what should be done	Records what was actually done
Specifies formula/process	Records actual quantities/process details
General reference for manufacturing	Batch-specific
Used for each applicable batch	Identifies individual batch execution

Quick Summary

- It is linked to **Rules 74, 74A, 74B, 78 and 78A**.
- Manufacturing records are **batch-specific**.
- The record must refer to the **Master Formula Record**.
- All ingredients and actual quantities must be documented.
- Raw-material **control numbers** must be recorded.
- **In-process controls** must be documented.
- Analytical report results must be linked to the batch record.
- Batch release must be verified and documented.
- Records must be maintained for the period prescribed under the relevant licensing provision.

SCHEDULE V

Schedule V prescribes **standards for Patent or Proprietary Medicines (P&Ps) u**nder **Rule 124.**

Standards for Patent or Proprietary Medicines Containing Vitamins

It states that patent or proprietary medicines containing vitamins for:

- ❖ **prophylactic use**
- ❖ **therapeutic use**
- ❖ **paediatric use**

shall contain the vitamins in quantities **not less than and not more than those specified in the Schedule**, when administered in a single daily dose or in two divided daily doses.

Therefore: **Schedule V controls both under-strength and excessive vitamin content.**

General Standards for Different Categories

The Schedule states that when a pharmaceutical product contains **several active ingredients**, their selection must be such that:

- The ingredients **do not interact with each other**.
- They do not adversely affect the **safety** of the product.
- They do not adversely affect **therapeutic efficacy**.
- The combination does not create analytical difficulties in separately assaying the ingredients.
- Added substances/additives must be **innocuous**.
- Additives must not adversely affect the safety or therapeutic efficacy of active ingredients.
- Additives must not interfere with **assay and identity tests** at the amount present.

General Dosage-Form Requirements

Schedule V states that patent or proprietary medicines must comply with the **general requirements of the dosage form** given in the **Indian Pharmacopoeia**.

If the dosage form is not included in the Indian Pharmacopoeia but is included in another pharmacopoeia prescribed for the purposes of the Second Schedule, the dosage form must comply with the general requirements of that pharmacopoeia.

The Schedule specifically mentions that general requirements include compliance with:

- **colour consistency**
- **clarity**
- **stability**
- freedom from contamination with **foreign matter**
- freedom from **fungal growth**
- absence of **chipping and capping of tablets**
- absence of **cracking of coating**
- absence of **mottled appearance**
- absence of other characteristic defects detectable by visual inspection.

Standards for Tablets

Tablets must:

- comply with the requirements for tablets laid down in the **Indian Pharmacopoeia**;
- indicate the **nature of coating on the label**;
- declare permitted colours when added;
- declare the nature of the tablet, such as:
  - uncoated;
  - sugar-coated;
  - film-coated

Standards for Capsules

Schedule V requires medicines in capsule dosage form to comply with the requirements for capsules laid down in the **Indian Pharmacopoeia**.

In addition, the capsules should be free from:

- distortion of shape;
- discolouration;
- leakage of powder from joints;
- pinholes;
- cracks;
- other physical defects.

Standards for Liquid Oral Dosage Forms

Parameter	Schedule V requirement
Emulsions	Must disperse uniformly on shaking
Suspensions	Must disperse uniformly on shaking
Homogeneous solutions	Must contain no sediment
Net volume	Must not be less than labelled volume
Ethanol content	90–110% of labelled content

SCHEDULE N

List of Minimum Equipment for the Efficient Running of a Pharmacy and is associated with **Rule 64(1)**.

The Schedule covers four broad areas:

1. **Entrance**
2. **Premises**
3. **Furniture and apparatus**
4. **General provisions**

It also provides a detailed list of the **minimum apparatus and books** required for making official preparations and prescriptions.

Overall Structure of Schedule N

Section	Major requirements
Entrance	Pharmacy inscription
Premises	Construction, size, lighting, ventilation, water, flooring, walls, dispensing area
Furniture & apparatus	Storage, containers, dispensing bench, poison cupboard, labels, concentrated solutions
Minimum apparatus	Balances, beakers, bottles, mortars, measures, pipettes, etc.
Books	IP, NFI, Drugs & Cosmetics Act/Rules, Pharmacy Act, Dangerous Drugs Act
General provisions	Registered pharmacist, cleanliness, records, poison storage, labelling
Licensing Authority	May relax or impose additional requirements in particular cases

1. Entrance: The front of a pharmacy must bear an inscription, **"Pharmacy"".**

2. Premises

2.1. Separation from private rooms

2.2. Construction and Environmental Requirements

The pharmacy premises must be:

- well built;
- dry;
- well lit;
- ventilated; and
- of sufficient dimensions for proper storage and handling of goods.

2.3. Storage of Medicaments and Poisons

The premises must be sufficiently large to allow goods in stock, particularly **medicaments & poisons** to be kept in a **clearly visible and appropriate manner.**

2.4. Minimum Dispensing Department Area

The dispensing department must have an area of minimum 6 m^2 for one pharmacist and for each additional pharmacist additional 2 m^2.

2.5. Height of Pharmacy Premises: The height of the premises must be at least **2.5 metres.**

2.6. Flooring Requirements: Smooth & Washable to facilitates cleaning, maintenance of hygiene and removal of dust and contaminants.

2.7. Wall Requirements: The walls must be plastered; or tiled; or oil painted so that they provide a smooth, durable and washable surface and must be free from holes, cracks, crevices.

2.8. Water Supply: A pharmacy must have ample supply of good quality water.

2.9. Dispensing Department Barrier: The dispensing department must be separated by a barrier to prevent admission of the public.

3. Furniture and Apparatus: Schedule N states that furniture and apparatus should be adapted to the uses for which they are intended.

3.1. Storage of Drugs, Chemicals and Medicaments: Must be kept in a room appropriate to their properties and in suitable special containers.

3.2. Containers and Protection from Dust

Drawers, glasses and other containers used for medicaments must:

- be of suitable size;
- be capable of being closed tightly; and
- prevent entry of dust.

3.3. Labelling of Containers

Every container must bear a label of appropriate size, easily readable; and bearing the names of medicaments as given in the **Pharmacopoeias**.

3.4. Dispensing Bench: A pharmacy must have a **dispensing bench** whose top must be covered with a **washable and impervious material**

3.5. Poison Cupboard: A pharmacy must have a **cupboard with lock and key** for storage of poisons. It must be clearly marked **"POISON".**

3.6. Concentrated Solutions: Containers of all concentrated solutions must bear a special label or be marked **"To be diluted".**

4. Minimum Apparatus

Schedule N then specifies the minimum apparatus required for making official preparations and prescriptions.

Apparatus Examples with Specifications

Equipment	Specification
Dispensing balance	Sensitivity 30 mg
Counter balance	Capacity 3 kg; sensitivity 1 g
Measuring glass	10, 25, 100, 500 mL
Graduated pipettes	2, 5, 10 mL
Thermometer	0–200°C
Metric weights	1 mg–100 g

5. Books Required in a Pharmacy

Schedule N also specifies the books/reference texts that should be available.

S. No.	Book/legislation
1	The Indian Pharmacopoeia – Current Edition
2	National Formulary of India – Current Edition
3	The Drugs and Cosmetics Act, 1940
4	The Drugs and Cosmetics Rules, 1945
5	The Pharmacy Act, 1948
6	The Dangerous Drugs Act, 1930

6. General Provisions

6.1. Continuous Personal Supervision

A pharmacy must be conducted under the continuous personal supervision of a Registered Pharmacist and the pharmacist's name must be displayed conspicuously on the premises.

6.2 Dress of the Pharmacist

The pharmacist must always put on clean white overalls.

6.3. Cleanliness and Order

The pharmacy premises and fittings must be properly maintained in good order and clean.

6.4. Records and Registers

All records and registers must be maintained **in** accordance with the laws in force.

6.5. Handling of Poison Cupboard

Whenever a container is taken from the poison cupboard, It must be replaced immediately after use. The poison cupboard must then be locked. The keys must remain in the personal custody of the responsible person.

6.6. Labelling of Dispensed Medicaments

Dispensing does not end with preparation, the product must also be appropriately labelled before supply.

7. Discretion of the Licensing Authority: The requirements of Schedule N are subject to modification at the discretion of the Licensing Authority. The Licensing Authority may relax requirements or impose additional requirements.

SCHEDULE T

Good Manufacturing Practices for Ayurvedic, Siddha and Unani Medicines under **Rule 157.**

Schedule T applies to manufacturers of **Ayurvedic medicines, Siddha medicines and Unani medicines.**

The document also provides a specific exemption:

Registered **Vaidyas, Siddhas and Hakeems** who prepare medicines themselves **for dispensing to their own patients and do not sell the medicines in the market** are exempted from the GMP requirements.

OBJECTIVES:

1. Raw materials are **authentic**.
2. Raw materials are of **prescribed quality**.
3. Raw materials are **free from contamination**.
4. Manufacturing is carried out according to the **prescribed process**.
5. Adequate **quality-control measures** are adopted.
6. The drug released for sale is of **acceptable quality**.
7. The manufacturer's procedures and methodology are **documented** and available for reference and inspection.

Basic Structure of Schedule T

Part	Subject
Part I	Good Manufacturing Practices
Part II-A	Machinery, equipment and minimum manufacturing premises for various Ayurveda/Siddha categories
Part II-B	Machinery, equipment and minimum manufacturing premises for Unani medicines
Part II-C	Equipment recommended for in-house Quality Control
Part D	Supplementary GMP guidelines for Rasaushadhies/Rasamarunthukal and Kushtajat

PART I – GOOD MANUFACTURING PRACTICES

1. Factory Premises

Area	Purpose
Raw-material receiving and storage	Receipt and storage of raw materials
Manufacturing process areas	Manufacturing operations
Quality-control section	Testing and quality control
Finished-goods store	Storage of approved finished products
Office	Administrative activities
Rejected goods/drugs store	Segregated storage of rejected material.

2. Location and Surroundings

The factory should be situated and constructed in a manner that prevents contamination from:

- open sewerage;
- drains;
- public lavatories;
- factories producing disagreeable or obnoxious odours/fumes
- excessive soot;
- dust;
- smoke.

3. Building Requirements

The building should permit production under **hygienic conditions**.

It should:

- be free from cobwebs;
- be free from insects and rodents;
- have adequate light;
- have adequate ventilation;
- have floors and walls that are not damp or moist.

4. Working Space and Prevention of Mix-ups

Adequate working space should be provided for:

- orderly placement of equipment;
- orderly placement of materials;
- preventing mix-ups between different drugs/components;
- controlling cross-contamination;
- avoiding omission of manufacturing/control steps.

5. Water Supply & Waste Disposal

- Adequate water must be available of pure and potable quality for manufacturing process.
- Waste water and residues originating from manufacturing sections, Laboratories which could be prejudicial to workers or public health must be appropriately disposed of.

6. Stores

- Have proper ventilation.
- Be free from dampness.
- Provide independent and adequate space for different categories of materials.

Separate storage should be provided for:

- Raw materials
- Packaging materials
- Finished products

6.1. Storage of Raw materials

Aspect	Requirement
Storage location	All raw materials must be stored in the Raw-Material Store
Container selection	Containers should be selected based on experience, characteristics of the raw material, and requirements of the Ayurveda/Siddha/Unani system
Protection required	Containers must protect raw materials from dampness, microbiological contamination, rodents, insects and damage
Special storage conditions	Raw-material store may be subdivided/enclosed/cabinized where special environmental conditions are required

Categories of Raw Materials

- Metallic-origin raw materials
- Mineral-origin raw materials
- Animal-source raw materials
- Fresh herbs
- Dry herbs/plant parts
- Excipients
- Volatile oils, perfumes & flavours
- Plant concentrates, extracts & exudates/resins

Identification & Labelling of Raw Materials

Label/Identification Requirement	Details
Name	Name of raw material
Source	Source of supply
Material status	**UNDER TEST / APPROVED / REJECTED**
Batch identification	Batch No. / Lot No.
Receipt information	Date of receipt

Testing, Approval & Handling of Raw Materials

Stage	Requirement
Sampling	All raw materials must be sampled
Testing	By in-house Ayurvedic/Siddha/Unani experts/QC technical personnel or Government-approved laboratories
Approval	Materials may be used only after satisfactory testing and approval
Rejected materials	Remove from main raw-material store and keep in a separate room
Stock rotation	Follow FIFO (First-In-First-Out) wherever necessary
Records	Maintain records of receipt, testing, approval/rejection and use

6.2. Packaging Materials

Aspect	Requirement
Examples	Bottles, jars, capsules and other packaging materials
Storage	Packaging materials must be **properly stored**
Containers & closures	Must be **adequately cleaned and dried** before packing
Purpose	Prevent contamination and ensure suitability for product packing

6.3. Finished-Goods Store

Stage	Requirement/Action
Transfer from production	Properly packaged finished products are transferred to the Finished-Goods Store
Initial status	Products are placed in an area marked “QUARANTINE”
Quality verification	Checked by Quality-Control Laboratory and Experts
Checks performed	Packing, labelling and finished-product quality
Approval	After satisfactory verification, products are transferred to “APPROVED FINISHED GOODS STOCK”
Dispatch	Only approved finished goods may be dispatched
Distribution	Distribution records must be maintained
Special storage	Necessary environmental/storage conditions must be provided where required

7. Health, Clothing, Sanitation and Hygiene of Workers

- All workers must be free from contagious diseases.
- Workers must wear suitable clothing based on nature of work & Climate.
- Clothing should be clean.
- Where required, uniforms should include coverings for hands, Feet & Head.
- Facilities for personal cleanliness should include clean towels, Soap & Scrubbing brushes.
- Separate lavatories must be provided for men & women.
- Lavatories should be located away from processing rooms.
- Workers must also have facilities for changing clothes & keeping personal belongings

8. Medical Services

The manufacturer must provide **First-aid facilities & Medical examination**

Workers must undergo:

- Medical examination at employment
- Periodic medical examination thereafter

The periodic examination should be conducted **once every year** by a physician, with particular attention to freedom from infections.

Records of medical examinations must be maintained.

9. Machinery and Equipment

- Suitable equipment must be available according to Size of operation & Nature of product
- Equipment may be; Manually operated, Semi-automatic, Electrical/steam operated or Fully automatic.
- Machines may be required for Crushing, Grinding, Powdering , Boiling, Mashing, Burning, Filtering, Drying, Filling etc.
- Adequate space must be provided between machines or rows of machines to ensure easy movement of workers & Orderliness of operations.
- Equipment must be; Properly installed, Properly maintained & Properly cleaned.

NOTE:

SOPs must be established for:

- Cleaning of each machine
- Maintenance of each machine
- Performance/operation of each machine

10. Batch Manufacturing Records (BMR)

The licensee must maintain a **BMR** for **every batch** of Ayurvedic, Siddha and Unani drugs.

This applies irrespective of whether the product is a classical preparation or patent/proprietary medicine.

The manufacturing record should include:

Raw-material information

- List of raw materials
- Quantities obtained from the store

In-process testing

Tests conducted during different manufacturing stages, including where applicable:

- Taste
- Colour
- Physical characteristics
- Chemical tests

Tests may be conducted on:

- Raw materials
- In-process materials
- Finished products

The records must be signed by:

- Production personnel
- Quality-control personnel

Finished-product transfer records

Records must also contain:

- Date of transfer to finished-goods store
- Quantity transferred
- Finished-product testing
- Packaging records

Only products that have been **verified and accepted as being of appropriate quality** may be cleared for sale.

The manufacturer should also maintain records of:

- Date
- Manpower used
- Machines used
- Equipment used
- **Various in-process operations such as:**
  - Shodhana
  - Bhavana
  - Burning
  - Fire processes
  - Specific grinding processes

11. Distribution Records

Records of sale and distribution must be maintained for **each batch**. The purpose is to facilitate **Prompt and complete recall** if required.

For ordinary products, records should be maintained up to the **expiry date of the batch**.

However, certain Ayurvedic/Siddha/Unani medicines mentioned in Schedule T do not have an expiry date because their efficacy is considered to increase with time.

Examples include Bhasma, Rasa, Sindura, Karpu/Uppu/Puram, Kushta , Asava-Arishta.

For such products, records should be maintained for **five years** after exhaustion of stock.

12. Records of Market Complaints

Manufacturers must maintain a register for all market complaints relating to products sold.

The register must contain:

- Market complaint details
- Investigation carried out
- Corrective action initiated
- Measures to prevent recurrence

The manufacturer must submit the record of complaints to the Licensing Authority, **once every six months**

The register must also be available for inspection.

Adverse reactions

A separate register must be maintained for adverse reactions associated with Ayurvedic, Siddha and Unani drugs.

The manufacturer must investigate:

- Whether the reaction was due to a product defect
- Whether the reaction has already been reported in literature
- Whether it represents a new observation

13. Quality Control

Every licensee must provide a **quality-control facility** either:

1. Within the manufacturing premises, **or**
2. Through a Government-approved testing laboratory.

Testing must follow:

- Ayurveda, Siddha and Unani pharmacopoeial standards.

Responsibilities of Quality Control

- Verify raw materials.
- Monitor in-process quality checks.
- Control the quality of finished products released to the finished-goods store/warehouse.

Minimum requirements of Quality-Control Section Area

A minimum area of **150 sq. feet** is specified.

Reference materials

For identification of raw drugs:

- Reference books
- Reference samples

should be maintained.

Records

Manufacturing records should be maintained for various processes.

Controlled samples

Controlled samples of finished products from every batch should be retained until:

- Expiry date of the product, or
- **3 years**

as stated in the Schedule.

Storage monitoring

QC should supervise and monitor storage conditions for:

- Raw materials
- Semi-finished products
- Finished products

Shelf life

Records should be maintained for establishing:

- Shelf life
- Storage requirements

QC Personnel

Personnel	Minimum qualification specified
Ayurveda/Siddha/Unani expert	Degree qualification recognized under Schedule II of the Indian Medicine Central Council Act, 1970
Chemist	At least Bachelor's degree in Science, Pharmacy or Pharmacy (Ayurveda) from a recognized University
Botanist/Pharmacognosist	At least Bachelor's degree in Science (Medical), Pharmacy or Pharmacy (Ayurveda) from a recognized University

Requirements for Sterile Products

A. Manufacturing Areas

- Separate enclosed manufacturing areas must be provided.
- These areas must be specifically designed for sterile manufacturing.
- Air locks must be provided at entry.
- Areas should essentially be dust-free.
- Adequate air supply/ventilation must be provided.

For aseptic manufacture:

- Air must pass through **bacteria-retaining filters/HEPA filters**.
- Air pressure should be higher than adjacent areas.
- HEPA filter performance must be checked
- Records of filter checks must be maintained.

Surfaces in sterile manufacturing areas must facilitate:

- Cleaning
- Disinfection

Routine microbial counts must be carried out during manufacturing operations.

Results should be compared with established in-house standards and recorded.

Access control

Access to sterile manufacturing areas must be restricted to the minimum number of authorized personnel.

The procedure for entering or leaving the manufacturing area must be written down & Displayed.

Prevention of mix-ups and contamination

For products intended for terminal sterilization, the design must prevent mixing of:

- Products awaiting sterilization
- Products already sterilized

For terminally sterilized products, the layout must also prevent mix-ups between non-sterile products.

Specific precautions

Schedule T specifies measures including:

1. Separate block or adequately isolated enclosure.
2. Appropriate pressure differential.
3. Suitable exhaust system.
4. Laminar-flow sterile air system where required.
5. Checking and recording germicidal efficiency of UV lamps.
6. Examination of individual liquid/ophthalmic containers against black-white background using diffused light.
7. Checking actual yield against theoretical yield before final distribution.
8. Recording process controls such as:
    1. Room temperature
    2. Relative humidity
    3. Volume filled
    4. Leakage
    5. Clarity

PART II-A. Recommended Machinery, Equipment and Minimum Manufacturing Premises

An important provision is that:

- A machine specified for one category may also be used for another category.
- Certain manufacturing areas can be shared, including: Powdering, Furnace, Liquid packing, Avaleha and Paks

General manufacturing area

The table specifies **1200 sq. ft. covered area** with separate cabins or partitions for each activity.

Where Unani medicines are also manufactured in the same premises, an **additional 400 sq. ft.** is required.

Category	Minimum area	Major recommended equipment
Anjana/Pisti	100 sq ft	Karel, mechanized/motorized karel, end runner/ball mill, sieves/shifter
Churna/Nasya/Manjan/Lepa/Kwath Churna	200 sq ft	Grinder, disintegrator, pulverizer, powder mixer, sieves/shifter
Pills/Vati/Gutika/Matrikai/Tablets	100 sq ft	Ball mill, mass/powder mixer, granulator, drier, tablet machine, pill/vati cutter, coating/polishing pan
Kupi-pakava/Ksara/Parpati/Lavana Bhasma/Satva/Sindura/Karpu/Uppu/Param	150 sq ft	Bhatti, Karahi, SS vessels, Patila flask, Gaj Put Bhatti, muffle furnace, end/edge runner, exhaust fan
Kajal	100 sq ft	Earthen lamps, triple roller mill, end runner, sieves, SS Patila, filling/packing equipment, exhaust fan, UV lamps
Capsules	100 sq ft	Air conditioner, dehumidifier, hygrometer, thermometer, capsule filling machine, chemical balance
Ointment/Marham/Pasai	100 sq ft	Tube filling machine, crimping machine, ointment mixer, end runner/mill, SS storage container, SS Patila

PART II-A. Recommended Machinery, Equipment and Minimum Manufacturing Premises

Category	Minimum area	Major equipment
Pak/Avaleh/Khand/Modak/Lakayam	100 sq ft	Bhatti with exhaust and fly-proof arrangement, iron Kadahi/SS Patila, SS storage container
Panak/Syrup/Pravahi Kwath/Manapaku	150 sq ft	Tincture press, exhaust, Bhatti, bottle washing machine, filter press/gravity filter, liquid filling machine, capping machine
Asava/Arishta	200 sq ft	Fermentation tanks, containers, distillation plant where necessary, filter press
Sura	100 sq ft	Distillation plant, transfer pump, etc.
Ark/Tinir	100 sq ft	Maceration tank, distillation plant, liquid filling tank, gravity filter/filter press, visual inspection box
Tail/Ghrit/Ney	100 sq ft	Bhatti, Kadahi/SS Patila, SS storage containers, filtration equipment, filling tank/liquid filling machine
Aschyotan/Netra Malham/Panir/Karn Bindu/Nasabindu	100 sq ft	Hot-air oven, kettle, mixing arrangement, ointment mill, tube filling equipment, SS tanks, filters, liquid filling equipment, autoclave
Bhatti/Furnace/Boiler/ Puta area	200 sq ft	Separate area with ventilation, smoke removal and protection from flies, insects and dust

PART II-B-Unani Medicines

Category	Minimum area	Important equipment
Itrifal/Tirya/Majoon/Laooq/Jawarish/Khamiras	100 sq ft	Grinder/pulverizer, sieves, powder mixer, SS Patilas, Bhatti, plant mixer for Khamiras
Arq	100 sq ft	Distillation plant, SS storage tank, boiling vessel, gravity filter, bottle filling/washing/drying equipment
Habb/Pills/Tablets	100 sq ft	Ball mill, mass/powder mixer, granulator, drier, tablet machine, pill cutter, coating/polishing equipment
Sufoof/Powder	200 sq ft	Grinder/pulverizer, sieves, trays, scoops, powder mixer
Raughan/Oils	100 sq ft	Oil expeller, SS Patilas, oil filter, bottle filling machine, bottle drier, Bhatti

Shiyaf/Surma/Kajal	100 sq ft	End runner, mixing SS vessel
Marham/Zimad	100 sq ft	Kharal, Bhatti, end runner, grinder, pulverizer, triple roller mill where required
Qurs/Tablets	100 sq ft	Grinder/pulverizer, sieves, powder mixer, granulator, drier, tablet machine, dies/punches, coating/polishing equipment
Kushta	100 sq ft	Bhatti, Kharal, Sil Batta, earthen pots
Murabba	100 sq ft	Aluminium vessels, Gendna, Bhatti
Capsule	100 sq ft	Pulverizer, powder mixer, capsule-filling machine, AC, dehumidifier, balance, storage containers
Sharbat/Joshanda	100 sq ft	Tincture press, exhaust, Bhatti, bottle washer, filter press/gravity filter, filling equipment, hot-air oven, kettle
Qutoor-e-Chashm/Marham (eye preparations)	100 sq ft	Hot-air oven, thermostatically controlled equipment and kettle
Bhatti/Furnace/Boiler/Puta	200 sq ft	Separate, ventilated area with smoke removal and protection against contamination

PART II-C-Equipment for In-house Quality Control

Schedule T provides a list of recommended equipment for the **in-house QC section**.

It also specifically states that the manufacturing unit may alternatively have testing performed by a **Government-approved laboratory**.

Chemistry section

The listed equipment includes:

1. Alcohol determination apparatus
2. Volatile-oil determination apparatus
3. Boiling-point determination apparatus
4. Melting-point determination apparatus
5. Refractometer
6. Polarimeter
7. Viscometer
8. Tablet disintegration apparatus
9. Moisture meter
10. Muffle furnace
11. Electronic balance
12. Magnetic stirrer
13. Hot-air oven
14. Refrigerator
15. Glass/steel distillation apparatus
16. LPG gas cylinders with burners

1. Temperature-controlled water bath
2. Heating mantles/hot plates
3. TLC apparatus with accessories
4. Paper chromatography apparatus
5. Sieves, size 10–120, with sieve shaker
6. Centrifuge machine
7. Dehumidifier
8. pH meter
9. Limit-test apparatus

Pharmacognosy section

The listed facilities/equipment include:

1. Binocular microscope
2. Dissecting microscope
3. Microtome
4. Physical balance
5. Aluminium slide trays
6. Stage micrometer
7. Camera lucida -prism and mirror type
8. Chemicals, glassware, etc.

PART II-D- Supplementary GMP for Rasaushadhis/ Rasamarunthukal / Kushtajat

These supplementary guidelines apply to **herbo-mineral-metallic formulations** of Ayurveda, Siddha and Unani medicines.

They cover preparations such as:

- Bhasma
- Sindura
- Pishti
- Kajjali
- Khalviya Ras
- Kupipakwa
- Rasayan
- Parpati
- Potali Rasa
- Satwa of metals/minerals
- Druti Parpam
- Karpu
- Kushta

Objectives

- Authenticity of raw drugs, minerals and metals
- In-process validation
- Quality-control parameters
- Compliance with classical texts
- Safety measures during manufacturing.

Specific manufacturing areas for Rasaushadhis

Area	Minimum requirement
Bhatti/heating section	100 sq ft
Grinding/drying/processing section	100 sq ft
Rasaushadhi-related store	100 sq ft

The Bhatti section should have:

- Proper ventilation
- Exhaust
- Chimney where appropriate

Grinding and drying section:

The grinding/drying/processing section may use:

- Manual equipment
- Mechanical equipment
- Oven

Drying should be performed in an area covered with:

- Glass
- Other transparent material

to permit sunlight where applicable.

Where an oven is used, the temperature may be selected according to the material and purpose of drying.

Safety for toxic fumes

Special attention must be given to products generating toxic fumes such as SO_2, Arsenic vapours & Mercury vapours

Suitable:

- Ventilation, Air-exhaust systems, Chimneys, Ducts & Scrubbing systems

should be provided to remove fumes and smoke and protect:

- Personnel
- Environment

Cleaning of Rasaushadhi equipment

Because processing may introduce:

- Heavy-metal contamination
- Cross-contamination

equipment cleaning is particularly important.

Equipment should be cleaned after **every process** using an appropriate cleaning agent that:

- Does not react with the equipment material.
- Does not have undesirable properties such as corrosiveness.

Temperature records

Records must be maintained for temperatures achieved throughout the **Bhasmikaran** process.

This applies when using:

- Classical Puta
- Oil-fired furnaces
- Gas-fired furnaces
- Electrically operated furnaces

Suitable temperature-measuring instruments may include:

- Pyrometer
- Pyrograph
- Heat sensors connected to a computer

Aluminium or aluminium-alloy equipment should **not** be used for hand-operated extruders used for large quantities.

Rasaushadhi Quality-Control Records

A. Shodhan Register

The register should contain:

1. Serial number
2. Batch number and batch size
3. Date, time and duration
4. Raw material name, quality reference and quantity
5. Quantity of Shodhana Dravya
6. Book reference followed
7. Methodology

B. Bhavana and Puta Register

Should contain:

1. Serial number
2. Batch number
3. Date and time
4. Starting material and quantity
5. Quantity of Nirvapya Dravya
6. Quantity of Bhavana Dravya
7. Start/completion date and time and duration of Bhavana/Mardana
8. Type and number of Puttas
9. Date and time of completion of Puttas
10. Colour and texture/product standards
11. In-process tests, including Bhasma Pariksha and other tests
12. Temperature attainment records where a specific temperature is required.

C. Grinding Record Register

1. Serial number
2. Batch number
3. Date and time
4. Material name and quantity
5. Equipment used — SS/granite
6. Grinding duration
7. Number of repeated grinding operations, where required.

D. Packing details

The record should contain:

1. Name of Rasaushadhi
2. Dosage form- e.g. powder, pill, tablet
3. Weight of Rasaushadhi in each unit

Product Quality Control of Rasaushadhis

Quality assurance depends heavily on using:

- Pharmacopoeial-quality metals
- Pharmacopoeial-quality minerals

The manufacturer must establish **in-house standards** to ensure uniform quality.

Quality testing includes characteristics such as:

- Colour
- Taste
- Varitaratwa
- Rekhapurnatwa
- Laghutva
- Nirdhumatwa
- Dntagre Kachakacha
- Niruttha
- Apunarbhava
- Nischandratwa

Additional testing includes:

- Particle size
- Physicochemical characterization
- Standard manufacturing process
- Disintegration time of pills/vati/tablets

Product Recall

Literature inserted inside the product package should provide:

- Name of manufacturing unit
- Address
- Telephone number

for reporting adverse drug reactions by:

- Physicians
- Patients

When an ADR is reported, the manufacturer is responsible for ensuring **recall of the product from the market**.

SOPs must specify storage of recalled Rasaushadhis in a Secure & Segregated area until final disposal.

Medical examination of Rasaushadhi manufacturing employees

Employees involved in Rasaushadhi manufacturing must undergo medical examination a**t least once every year**

The purpose is to identify adverse effects of the manufacturing materials/processes.

Necessary investigations should be performed to ensure there is no effect on vital organs.

Annual examination reports must be available to statutory inspectors during GMP inspections.

Self-Inspection

Release of Rasaushadhis should be controlled by a person trained in:

- Specific processing features
- Quality assurance of Rasaushadhis

Personnel working in production and quality assurance should have adequate training in Rasaushadhi manufacturing.

The specified qualification is at least:

- Degree in Ayurveda/Siddha/Unani medicines, **or**
- B.Pharm degree in Ayurvedic/Siddha/Unani medicines.

Dosage Forms of Rasaushadhis

Rasaushadhis may be prepared into acceptable dosage forms such as churna, vati, guti, tablets & capsules.

Suitable permissible fillers or binding agents may be used according to the relevant pharmacopoeial requirements.

The label must indicate the quantity of ASU medicine present in each tablet, pill & capsule in addition to the quantity of filler.

Rasaushadhis must be packed in a dosage form **ready for consumer use**.

Grinding and weighing of individual doses of potentially poisonous products is not permitted in patient-consumer packs because dose variation may contribute to adverse drug reactions.

For hospital bulk packs, this restriction does not apply, but the label must clearly indicate **“Hospital pack”.**

SCHEDULE Y: Requirements & Guidelines for Permission to Import/Manufacture New Drugs or Undertake Clinical Trials

Schedule Y applies to

Area	Purpose
Import of new drugs	Permission for import
Manufacture of new drugs	Permission for manufacture for sale
Clinical trials	Permission to undertake clinical trials
New drug applications	Submission and evaluation of required data
Safety & efficacy	Establish evidence before marketing
Ethical conduct	Protection of clinical-trial subjects

Associated Rules: 122A, 122B, 122D, 122DA, 122DAA and 122E.

Application for Permission

Application must be made in **FORM 44.**

The application should be accompanied by data specified under the relevant appendices of Schedule Y.

Major category of data
• Chemical & pharmaceutical information
• Animal pharmacology
• Animal toxicology
• Human/clinical pharmacology
• Clinical trial data
• Regulatory status in other countries
• Prescribing information
• Quality-control/testing information
• Samples of drug substance & finished product

Data Required for New Drug Application

Data category	Main information
Chemical & pharmaceutical	Identity, structure, physicochemical properties, formulation
Animal pharmacology	Specific & general pharmacological actions
Pharmacokinetics	Absorption, distribution, metabolism, excretion
Animal toxicology	Systemic, reproductive, local, genotoxicity, carcinogenicity etc.
Human pharmacology	Phase I data
Clinical efficacy/safety	Phase II & III data
Special populations	Geriatric, paediatric, pregnancy/nursing etc.

Regulatory status	Approval/restrictions/withdrawal in other countries
Prescribing information	Complete proposed package insert
QC information	Testing protocols, impurity profile, release specifications

Clinical Trial Requirements According to Origin of Drug

Situation	Requirement
New drug discovered in India	Clinical trials in India from Phase I
New drug discovered outside India	Phase I data from other countries submitted; permission may allow Phase I/II/III as specified
Before marketing in India	Phase III trials required in India for drugs discovered outside India, as stated in this Schedule
Multinational development	Number of Indian sites/subjects + justification to be submitted
Permission for phases	Generally given progressively based on data from earlier phases

Documents for Initiating a Clinical-Trial Phase

Document	Schedule Y reference
Investigator's Brochure	Required
Proposed Protocol	**Appendix X**
Case Record Form	Required
Informed Consent Documents	**Appendix V**
Investigator's Undertaking	**Appendix VII**
Ethics Committee clearance	**Appendix VIII**

****Clinical study reports submitted under relevant sections should follow the format in **Appendix II**.

Prescribing Information

*Generic name *Composition *Dosage form *Indications *Dose & method of administration *Use in special populations *Contraindications *Warnings *Precautions *Drug interactions *Undesirable effects *Overdose *Pharmacodynamic properties *Pharmacokinetic properties *Incompatibilities *Shelf-life *Packaging *Storage & handling instructions

Package inserts, promotional literature and patient-education material should remain consistent with the approved prescribing information`.

Special Provision for Serious/Life-threatening Diseases

For drugs indicated for:

- Life-threatening diseases
- Serious diseases
- Diseases of special relevance to Indian health

the Licensing Authority may consider **abbreviation, deferral or omission of toxicological/clinical data requirements**, as considered appropriate.

Clinical Trial: Basic Approval Requirements

A clinical trial can be initiated only after:

Licensing Authority permission

↓

Ethics Committee approval

↓

Site-specific approval

↓

Trial initiation

Investigators must possess appropriate:

- Qualifications
- Training
- Experience
- Access to appropriate facilities.

Protocol Amendments

Situation	Requirement
Amendment before/during trial	Notify Licensing Authority in writing
Protocol deviation/change	Prior written EC + Licensing Authority approval
Immediate hazard	Exception permitted to eliminate immediate hazard
Logistic/administrative change	Exception permitted
Immediate exceptions	Notify EC & Licensing Authority immediately
Administrative/logistic changes	Notify Licensing Authority within **30 days**

Responsibilities of Sponsor

*Establish & maintain Quality Assurance system *Ensure compliance with protocol *Ensure data generation/documentation/reporting *Follow GCP *Follow statutory requirements *Maintain documented SOPs *Submit trial status reports *Submit report for prematurely discontinued studies *Report serious adverse events *Provide medical management/financial compensation where applicable *Submit compensation details to Licensing Authority

Sponsor: Premature Discontinuation

Summary report: Within 3 months

It should contain information:

- Brief study description
- Number of patients exposed
- Dose
- Duration of exposure
- ADR details, if any
- Reason for discontinuation/non-pursuit of application

Responsibilities of Investigator

- Conduct trial according to protocol
- Follow GCP
- Maintain documented SOPs
- Ensure adequate medical care
- Report serious adverse events
- Provide informed-consent information
- Explain compensation rights
- Ensure qualified study personnel
- Maintain accurate records
- Cooperate with audit/inspection
- Maintain confidentiality
- Report protocol changes/unanticipated problems

Serious Adverse Event Reporting

Investigator within **24 hours** send report to Licensing Authority + Sponsor + Ethics Committee. After due analysis; Report forwarded within **14 days** to the specified authorities.

If the Investigator fails to report within the stipulated time, reasons for delay must be furnished

Informed Consent

Every trial requires:

- Information provided verbally.
- Patient Information Sheet
- Non-technical & understandable language
- Written Informed Consent Form
- EC approval of consent documents
- Documents furnished to Licensing Authority
- Amendments require EC approval before implementation

When Subject Cannot Give Consent?

Situation	Consent
Unconscious subject	Legally acceptable representative
Minor	Legally acceptable representative
Severe mental illness/disability	Legally acceptable representative
Subject/representative unable to read/write	**Impartial witness** throughout consent process

Ethics Committee: Main Responsibility

1. Safeguard rights, safety & well-being of trial subjects
2. Special attention to vulnerable groups such as:

- Prisoners
- Armed forces personnel
- Students/staff in academic institutions
- Patients with incurable diseases
- Economically disadvantaged persons
- Emergency patients
- Minors
- Homeless persons
- Refugees
- Persons unable to provide consent

Phases of Clinical Trials

Phase	Main purpose
Phase I	Safety & tolerability; initial human exposure
Phase II	Therapeutic effectiveness, short-term adverse effects, dose/regimen
Phase III	Confirm therapeutic benefit, safety & efficacy; support marketing approval
Phase IV	Post-marketing studies; optimize use, safety, interactions, epidemiology etc.

Phase I: Human Pharmacology

Objective	Purpose
Maximum tolerated dose	Determine tolerability & adverse reactions
Pharmacokinetics	ADME characterization
Pharmacodynamics	Drug effects in relation to exposure
Early drug activity	Preliminary evidence of activity/benefit

Usually conducted in healthy volunteers or selected patients; drugs with significant toxicity may be studied in patients.

Phase II: Therapeutic Exploratory Trials

- Evaluate effectiveness for particular indication
- Identify common short-term side effects
- Identify risks
- Determine dose/dose regimen for Phase III

Phase III: Therapeutic Confirmatory Trials

#Confirm therapeutic benefit #Confirm safety #Confirm efficacy #Establish dose-response relationships #Evaluate wider patient populations #Evaluate different stages of disease #Evaluate combination therapy #Provide adequate basis for marketing approval.

For drugs approved outside India, the Schedule states that Phase III studies in India are primarily intended to generate efficacy and safety evidence in Indian patients.

Phase IV: Post-Marketing Trials

- Drug-drug interaction studies
- Dose-response studies
- Safety studies
- Mortality/morbidity studies
- Epidemiological studies
- Studies supporting use under approved indications

These go beyond the safety, efficacy and dose information established before approval.

Studies in Special Populations

1. Paediatric Studies

Studies according to age group and safety considerations

For diseases predominantly/exclusively affecting children:

- Paediatric clinical data should be generated.
- Initial safety/tolerability may generally be obtained in adults unless inappropriate.
- Serious/life-threatening diseases with limited treatment options may require early paediatric inclusion

Consent: Parent/Legal Guardian consent + child information where appropriate

2. Pregnant & Nursing Women

Situation	Requirement
Drug intended for pregnant/nursing women	May be included where appropriate
Drug intended for foetus/infant	Appropriate inclusion
Non-pregnant data inadequate	Pregnancy/nursing studies may be required
Pregnancy follow-up	Pregnancy, foetus & child
Drug excretion in milk	Examine where applicable
Infant	Monitor predicted pharmacological effects

PSUR (Periodic Safety Update Reports) Schedule

Period after approval	Frequency
First 2 years	Every **6 months**
Subsequent 2 years	**Annually**
Licensing Authority	May extend requirement
PSUR submission	Within **30 calendar days** of reporting period
Serious unexpected ADR	Within **15 days** of initial receipt by applicant

PSUR Structure

- Title page
- Introduction
- Worldwide marketing authorisation status
- Safety-related regulatory actions
- Changes to reference safety information
- Estimated patient exposure
- Individual case histories
- Studies
- Other information
- Overall safety evaluation
- Conclusion
- Appendices

Bioavailability/Bioequivalence

- BE with reference formulation where applicable
- Conduct under labelled administration conditions
- Assess systemic absorption where relevant
- Evaluate food effect for oral administration
- Submit dissolution data for solid oral dosage forms
- Establish bioequivalence/bioavailability
- Establish dosage correlations where relevant
- Follow prescribed BA/BE guidelines

PRECLINICAL STUDIES

Animal studies	**Major areas**
Pharmacology	Specific effects
General pharmacology	General physiological effects
PK	ADME
Systemic toxicity	Toxic effects
Reproductive toxicity	Fertility/development
Local toxicity	Dermal/ocular/other
Genotoxicity	Genetic damage
Carcinogenicity	Tumour-forming potential
Allergenicity	Hypersensitivity

APPENDICES

Appendix	Topic
Appendix I	Data required for new drug application
Appendix IA	New drug already approved in the country
Appendix IB	Relevant data requirements
Appendix III	Animal toxicology
Appendix IV	Animal pharmacology
Appendix VI	Fixed Dose Combinations
Appendix VIII	Ethics Committee
Appendix IX	Stability testing
Appendix X	Clinical trial protocol
Appendix XI	SAE reporting
Appendix XII	Compensation

Fixed Dose Combinations (FDCs)

FDC category	Requirement
One or more ingredients are new drugs	Data similar to new drug requirements
Already approved ingredients combined for first time	Appropriate pharmaceutical/clinical data
Significant PK/PD interaction possible	Additional evidence required
FDC marketed abroad	Regulatory status/trial data may be submitted

Stability Testing

Purpose: Establish stability of drug/product during intended storage.

Important considerations:

- Storage conditions
- Container-closure system
- Stability-indicating parameters
- Product quality
- Proposed shelf life
- Clinical-trial duration where applicable

The Schedule specifically refers to Appendix IX for stability studies and requires stability information for the intended container-closure system in clinical-trial applications.

Schedule X: Drugs requiring special control under the Drugs and Cosmetics Rules, 1945

Schedule X = Strict control + documentation + secure storage

SCHEDULE X DRUGS

Broad category	Examples
CNS-active drugs	Certain hypnotics/sedatives and related agents
Antidepressant/antipsychotic agents	Selected substances
Potent/controlled substances	Selected substances
Other specified drugs	As listed in Schedule X

Prescription Requirements

Requirement	Schedule X
Prescription required?	**Yes**
Prescription from	Registered Medical Practitioner
Prescription	**In duplicate**
Dispensing	Controlled
Retention of prescription	**2 years**

Supply to Institutions

Schedule X drugs supplied to:

- Registered Medical Practitioners
- Hospitals
- Dispensaries
- Nursing Homes

must be supplied against a signed written order that must be preserved for 2 years.

Storage

Storage requirement	Purpose
Locked cupboard/drawer	Prevent unauthorised access
Reserved exclusively	Separate from ordinary drugs
Controlled access	Prevent misuse/diversion
Responsible handling	Accountability

Storage During Manufacturing

- Schedule X drugs should be stored in **bulk form**.
- When required at another location for manufacturing, they should be kept separately.
- They remain under the **direct custody of a responsible person**.

Record Keeping

Why are Records Important?

Records provide:

- Traceability
- Accountability
- Stock control
- Prevention of diversion
- Verification during inspection
- Facilitation of investigation

Records for Schedule X Raw Materials

Particular	Record
Date	Date of issue
Drug	Name
Opening stock	Quantity
Receipt	Quantity + source
Manufacturing use	Quantity used
Closing stock	Balance
Authentication	Signature of responsible person

Production Records

Particular	Record
Date	Date of manufacture
Drug	Name
Batch	Batch number
Raw material	Quantity used
Yield	Expected/anticipated yield
Actual yield	Actual quantity obtained
Wastage	Quantity wasted
Transfer	Quantity transferred to stock

Manufactured Drug Account

Particular	Record
Date	Date of manufacture
Drug	Name
Batch	Batch number
Opening balance	Quantity
Manufactured	Quantity
Sold	Quantity
Purchaser	Name & address
Closing balance	Balance
Authentication	Signature

Packing & Labelling

Dosage form	Maximum pack size
Tablets/Capsules	100 unit doses
Oral liquids	300 mL
Injections	5 mL

Hospital Packs

Condition	Requirement
Supply	Direct from manufacturer/distributor
Recipient	Hospital/dispensary
Retail sale	**Not permitted**
Supply to RMP	Hospital pack not to be supplied as retail

Dispensing of Schedule X Drugs

Requirement	Schedule X
Prescription	Required
Duplicate	Yes
Repeat dispensing	Only according to prescription
Prescription retention	2 years
Record keeping	Required
Substitution	Not permitted

What the Pharmacist Must Ensure?

Before dispensing:

Valid prescription
↓
Correct drug & quantity
↓
Verify prescriber details
↓
Make required record
↓
Dispense correctly
↓
Preserve prescription

Manufacturing License

Manufacturing Controls

- Maintain stock records
- Maintain production records
- Maintain sales/distribution records
- Secure storage
- Controlled movement
- Responsible-person custody
- Maintain required documentation
- Permit inspection and verification

Reference Samples & Inspection

The Rules also require licensees to maintain **reference samples from each batch** in specified quantities and periods, and an **Inspection Book in Form 35** is maintained for inspectors' observations.

Advertisement Restriction

Schedule X drugs cannot be advertised freely.

Advertisement requires: **Previous sanction of the Central Government**

Part XII-B of Schedule F

Aspect	Description
Main area	Blood Banks
Covers	Whole human blood and blood components
Purpose	To ensure safe collection, processing, storage and distribution of blood/blood components

Personnel Requirements

Personnel	Main responsibility
Medical Officer	Supervision of procurement, storage and cross-matching
Blood Bank Technician	Blood grouping, cross-matching and technical activities
Trained Laboratory Technician	May perform relevant activities when appropriately trained

Duties of Medical Officer

- Proper **procurement** of blood/blood components
- Proper **storage**
- **Cross-matching**
- Identification of **haemolysed blood**
- Ensuring that **expired blood or blood components are not supplied**

Blood Bank Technician

Requirement	Details
Qualification	Must possess the **qualification and experience specified in Part XII-B of Schedule F**
Alternative personnel	An **experienced Laboratory Technician** may be employed
Essential training	The alternative laboratory technician must be **trained in blood grouping and cross-matching**
Main competency	Should be capable of performing/assisting in blood-bank laboratory procedures, particularly **blood grouping and cross-matching**
Role in blood storage facility	Technical staff required for proper handling and storage of blood and/or blood components

Premises Requirements

The Blood Bank should have:

Requirement	Provision
Area	10 m²
Lighting	Well-lighted
Cleanliness	Clean premises
Air-conditioning	Preferably air-conditioned
Refrigeration	Appropriate blood-bank refrigerator
Temperature monitoring	Required
Alarm	Refrigerator fitted with alarm device

Blood Bank Refrigerator

- Appropriate capacity
- Alarm device
- Temperature indicator
- Regular temperature monitoring

Storage temperature: 2°C – 8°C

****Whole human blood → -2°C to 8°C**

Equipment Requirements for Blood Bank

Equipment / Facility	Purpose
Blood bank refrigerator	Storage of whole human blood at the specified temperature
Temperature indicator	Continuous/regular monitoring of refrigerator temperature
Alarm device	Alerts when the storage temperature moves outside the required range
Blood grouping equipment	Determination of **ABO and other blood groups**
Cross-matching equipment	Checking compatibility between **donor blood and recipient/patient blood**
Equipment for component preparation/storage	Required when blood components are prepared or stored
Other specialized equipment	Equipment appropriate to the specific blood components being handled

Procurement of Blood

Blood/blood components should be procured from **duly licensed sources**. For the specified FRU/CHC/PHC/hospital provisions, whole human blood and/or components are to be procured from:

- Government Blood Bank , Indian Red Cross Society Blood Bank, Regional Blood Transfusion Centre; provided these are duly licensed.

Testing of Blood

Testing is an essential safety requirement before use.

Blood should be tested for:

- **HIV I & II antibodies**
- **Hepatitis B surface antigen**
- **Malarial parasites**
- Other tests specified in the relevant **Indian Pharmacopoeia monograph**

Records & Registers

Blood Banks must maintain appropriate:

- Procurement records
- Blood/blood-component records
- Testing records
- Other records/registers required under **Part XII-B**

Storage of Samples

For the specified FRU/CHC/PHC/hospital provision:

Samples of:

- Donor's blood
- Patient's serum

are to be stored for **7 days after transfusion**

Special Provisions for FRU/CHC/PHC/Hospital

Requirement	Provision
Annual captive consumption	**Not more than 2,000 units**
Premises area	**10 m²**
Blood storage temperature	**2°C–8°C**
Sample retention	**7 days after transfusion**
Approval validity	**2 years**
Renewal application	**3 months before expiry**

Quality & Safety Responsibilities

Qualified personnel
↓
Proper procurement
↓
Testing
↓
Correct storage
↓
Cross-matching
↓
Identification of haemolysed/expired blood
↓
Proper documentation
↓
Safe blood for transfusion

DMR (OA): Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954

It came into force on **1 April 1955**. Its main purpose is to control objectionable advertisements relating to drugs and magic remedies.

- **The Act contains provisions dealing with prohibited advertisements, misleading advertisements, import/export of objectionable advertisements, penalties, inspection/search and offences by companies.**

Why Do We Need This Act?

A person may be persuaded by advertisements claiming that a product can:

- Cure serious diseases
- Improve sexual performance
- Correct menstrual disorders
- Prevent pregnancy or cause miscarriage
- Permanently cure chronic diseases
- Produce miraculous effects
- Treat diseases without medical consultation

Such claims can:

- Mislead patients
- Encourage self-medication
- Delay proper medical treatment
- Exploit vulnerable patients
- Promote unscientific or magical remedies
- Create false expectations regarding treatment

Objectives of the Act

- Control drug advertisements
- Protect the public
- Restrict advertisements for specified conditions
- Control magic remedies
- Control misleading claims
- Control import and export

DEFINITIONS

1. Advertisement

Advertisement includes any **notice, circular, label, wrapper, o**ther **document, a**ny announcement made **orally, a**ny announcement through; Light, Sound or Smoke.

The statutory definition specifically includes written, oral and certain transmitted forms of communication.

2. Magic Remedy

Magic remedy includes talisman, Mantra, Kavacha, any other charm, when it is alleged to possess **miraculous powers** for Diagnosis, Cure, Mitigation, Treatment, Prevention of disease or for affecting body structure, Influencing organic functions of the body

Taking Part in Publication?

The Act also covers persons who take part in publication of an advertisement as responsibility is not necessarily limited to the person who manufactures the drug.

This includes activities associated with:

- Printing the advertisement
- Publication of an advertisement outside the territories covered by the Act when done by or at the instance of a person residing within those territories

MAJOR PROHIBITIONS

Section	Subject
Section 3	Prohibition of advertisement of certain drugs for specified diseases/disorders
Section 4	Prohibition of misleading advertisements relating to drugs
Section 5	Prohibition of advertisement of magic remedies
Section 6	Prohibition of import/export of certain advertisements
Section 7	Penalties
Section 8	Powers of entry, search, etc.

Section 3: Prohibition of Certain Drug Advertisements

No person shall take part in publication of an advertisement referring to a drug in terms which suggest or are calculated to lead to its use for certain prohibited purposes.

There are **four major categories**.

- **Section 3(a)-**Miscarriage / prevention of conception
- **Section 3(b)-** Sexual pleasure
- **Section 3(c)-** Menstrual disorders
- **Section 3(d)-** Diseases/disorders/conditions specified

Example as per schedule J: **AIDS, Blindness, Diabetes, Deafness, Glaucoma, Goitre etc.**

Section 4: Misleading Advertisements

Section 4 prohibits publication of an advertisement relating to a drug if it contains matter which

4(a) Directly or indirectly gives a false impression regarding the true character of the drug

4(b) Makes a false claim for the drug

4(c) Is otherwise false or misleading in any material particular

Section 5: Magic Remedies

No person carrying on or purporting to carry on the profession of administering magic remedies shall take part in publication of an advertisement referring to a magic remedy that directly or indirectly claims efficacy for any purpose specified in **Section 3**.

Section 6: Import and Export

It prohibits the, **Import into India & Export from India** of documents containing advertisements referred to in Section 3, Section 4 & Section 5 whose import or export has been prohibited **under section 19 of the Sea Customs Act, 1878.**

Purpose is to prevent persons from bypassing the advertising restrictions by moving objectionable advertising material across borders.

Section 7: Penalty

First conviction punishable with:

- Imprisonment up to **6 months**, or **Fine**, or **Both.**

Subsequent conviction

- Imprisonment **up to 1 year** OR fine OR both

Section 8: Powers of Entry, Search, etc.

Any Gazetted Officer authorised by the **State Government** may**, *within the local limits of the area*** for which he is so authorised,

a) Enter and search at all reasonable times, with such assistants, if any, as he considers necessary, any place in which he has reason to believe that an offence under this Act has been or is being committed.
b) Seize any advertisement which he has reason to believe contravenes any of the provisions of this Act.
c) Examine any record, register, document or any other material object found in any place mentioned in clause (a) and seize the same if he has reason to believe that it may furnish evidence of the commission of an offence punishable under this Act.
d) Where any person seizes anything under clause (b) or clause (c), he shall, as soon as may be, inform a Magistrate and take his orders as to the custody thereof.

Other Sections of the Act

Section	Subject
Section 9	Offences by companies
Section 9A	Offences to be cognizable
Section 10	Jurisdiction to try offences
Section 10A	Forfeiture
Section 11	Officers to be deemed public servants
Section 12	Indemnity
Section 13	Other laws not affected
Section 14	Savings
Section 15	Power to exempt
Section 16	Power to make rules

Section 9: Offences by Companies

1. If the company contravening any of the provisions of this act, every person who, at the time the offence was committed, was in charge of, and was responsible to the company for the conduct of the business of the company as well as the company shall be deemed to be guilty of the contravention and shall be liable to be proceeded against and punished accordingly.

No such person liable to any punishment, if he proves that the offence was committed without his knowledge or that he exercised all due diligence to prevent the commission of such offence.

2. If a company breaks the law, the individual bosses (directors, managers, secretaries, or other officers) **cannot hide behind the company's identity**. If they helped commit the crime, allowed it to happen, or let it happen through neglect, they will be **personally charged, tried, and punished** alongside the company.

Section 9a: Offences to be cognizable

Cognizable offences are those in which the police can arrest the accused without a warrant. The police can also begin an investigation without the permission of the court. The accused is arrested and produced before the court at the stipulated time. According to Section 154 of the CrPC, a police officer is required to register an FIR in case of a cognizable offence.

Section 10: Jurisdiction to try offences

No court inferior to that of a Presidency Magistrate or a Magistrate of the first class shall try any offence punishable under this Act.

Section 10a: Forfeiture

Where a person has been convicted by any court for contravening any provision of this Act the court may direct that any document (including all copies), article or thing, in respect of which the contravention is made, including the contents thereof where such contents are seized, shall be forfeited to the Government.

Section 11: Officers to be deemed to be public servants

Every person authorised under section 8 shall be deemed to be a public servant within the meaning of section 21 of the Indian Penal Code (45 of 1860).

Section 12: Indemnity

No suit, prosecution or other legal proceeding shall lie against any person for anything which is in good faith done or intended to be done under this Act.

Section 13: Other laws not affected

The provisions of this Act are in addition to, and not in derogation of the provisions of any other law for the time being in force.

Section 14: Savings

Nothing in this Act shall apply to—

(a) Doctor Signboards: A qualified, registered medical doctor can put a simple sign or notice outside their clinic stating that they treat specific conditions listed in the Act. This allows patients to find the right care without the sign being treated as an illegal commercial advertisement.

(b) Scientific and Educational Books: Academic textbooks, research papers, and scientific or social treatises can freely discuss serious diseases and treatments from a genuine educational or scientific point of view.

(c) Confidential Medical Communications: Drug information or advertisements sent privately and confidentially *only* to registered medical practitioners (not to the general public) are permitted.

(d) Government Publications: Any informational material or advertisements about drugs printed, published, or authorized by the government are exempt.

(e) Older Sanctioned Ads: Advertisements that received official government approval *before* the 1963 amendment to the Act are allowed to continue. However, the government keeps the power to cancel these old permissions later if there is a valid reason, but only after giving the person a fair chance to explain why the permission should stay.

Section 15: Power to exempt from application of Act

If in the opinion of the Central Government public interest requires that the advertisement of any specified drug or class of drugs or any specified class of advertisements relating to drugs should be permitted, it may, by notification in the Official Gazette, direct that the provisions of sections 3, 4, 5 and 6 or any one of such provisions shall not apply or shall apply subject to such conditions as may be specified in the notification.

Section 16: Power to make rules.

The Central Government may, by notification in the Official Gazette, make rules for carrying out the purposes of this Act.
Such rules may

- specify any disease, disorder or condition to which the provisions of section 3 shall apply.
- Prescribe the manner in which advertisements of articles or things referred to in clause (c) of section 14 may be sent confidentially.
- Every rule made under this Act shall be laid, as soon as may be after it is made, before each of House of Parliament while it is in session for a total period of **thirty days** which may be comprised in one session or in two or more successive sessions and if before the expiry of the session in which it is so laid or the successive sessions aforesaid, both Houses agree in making any modification in the rule or both Houses agree that the rule should not be made, the rule shall thereafter have effect only in such modified form or be of no effect.

Sale of Drugs

Legal Framework for Sale of Drugs

Provision	Subject
Section 18	Prohibition of manufacture and sale of certain drugs/cosmetics
Section 18A	Disclosure of manufacturer
Section 18B	Maintenance of records & furnishing information
Rules 59–65	Licensing requirements and conditions
Rule 61	Forms of licences to sell drugs
Rule 62	Sale at more than one place
Rule 62A	Restricted licences
Rule 62B	Conditions before restricted licence
Rule 63	Duration/retention of licence
Rule 65	Conditions of licences

Types of Sale

Type	Meaning	Main requirement
Retail sale	Sale directly to patients/consumers	Qualified person/pharmacist where required
Wholesale sale	Sale to licensed dealers/institutions/hospitals, RMP's etc. for further supply/use	Competent person and appropriate licence
Restricted retail sale	Sale of drugs that do not require supervision of a qualified person	Restricted licence
Schedule X sale	Sale of Schedule X drugs	Special licence and stricter controls

Licence Forms

Rule 61 specifically groups licences according to whether the drugs belong to **Schedules C/C(1), X, or neither**.

Licence Form	Purpose
Form 20	Retail sale – drugs other than Schedules C, C(1) and X
Form 20A	Restricted retail sale – drugs other than Schedules C, C(1) and X
Form 20B	Wholesale – drugs other than Schedules C, C(1) and X
Form 21	Retail sale – Schedule C & C(1), excluding Schedule X
Form 21A	Restricted retail – specified Schedule C(1) drugs, excluding Schedule X
Form 21B	Wholesale – Schedule C & C(1), excluding Schedule X
Form 20F	Retail sale – Schedule X drugs
Form 20G	Wholesale – Schedule X drugs

Retail Sale of Drugs

Retail sale of drugs **other than those specified in Schedules C, C(1) and X**.

Important conditions include:

- Licence must be displayed prominently.
- Drugs must be purchased from a duly licensed dealer/manufacturer.
- Qualified person must supervise sale where required.
- Changes in qualified staff must be reported.
- Changes in constitution of the firm must be reported.

Retail sale of drugs specified in **Schedules C and C(1), excluding Schedule X**.

Under Rule 65; Certain drugs supplied on the prescription of a RMP must be supplied **by or under the personal supervision of a registered pharmacist and documented**.

Wholesale sale of drugs

Other than those specified in **Schedules C, C(1) and X**.

Conditions

- Drugs must be purchased from a duly licensed dealer/manufacturer.
- Sale generally must not be made for resale to an unlicensed person.
- Licence must be displayed.
- Changes in constitution of firm must be reported.

Drugs specified in **Schedules C and C(1), excluding Schedule X**.

The sale is to be made under the personal supervision of a competent person.

Records & Documentation

Feature	Retail	Wholesale
Purchase records	✓	✓
Sale cash/credit memo	✓	✓
Purchaser details	✓	✓
Licence number	Dealer's licence	Purchaser's licence where applicable
Batch number	✓	✓
Manufacturer	✓	✓
Competent person signature	Where applicable	Required on wholesale memo
Memo retention	As prescribed	**3 years** for wholesale memo

Restricted Licence

- A restricted licence is issued for sale of drugs where the sale does not require supervision of a qualified person.
- These licences are issued at the **discretion of the Licensing Authority**
- Restricted licences may be issued to; Dealers/persons dealing in drugs whose sale does not require supervision of a qualified person or Certain itinerant vendors in exceptional circumstances

Conditions Before Grant of Restricted Licence

Before granting Form 20A or 21A, the Licensing Authority considers:

1. Premises: Must be:

- Adequate
- Properly equipped
- Provided with appropriate storage accommodation

2. Applicant

Authority considers:

- Number of licences already granted in the locality
- Occupation/trade/business of applicant
- Previous conviction under the Act/Rules
- Previous suspension/cancellation of licence

The authority may refuse a licence where the applicant is considered **not a fit person**

Duration of Sale Licence

- Sale licences are generally **perpetual**, unless suspended or cancelled.
- However, **Licence Retention Fee** must be deposited **before expiry of every succeeding 5-year period**.
- Failure to pay can lead to **Retention fee + late fee** and ultimately **Deemed cancellation of licence.**

Offences

- Sale without valid licence
- Sale of adulterated drugs
- Sale of spurious drugs
- Sale of substandard drugs
- Sale in violation of licence conditions
- Sale of drugs in contravention of the Rules
- Failure to maintain required records
- Failure to disclose required information
- Sale of prescription-controlled drugs without valid prescription
- Sale to an unauthorised/unlicensed person where prohibited

Penalties

Major offence	Imprisonment	Fine
Adulterated/spurious drug likely to cause death/grievous hurt	10 years → life	≥ ₹10 lakh or 3× value
Adulterated drug / sale without valid licence	3 → 5 years	≥ ₹1 lakh or 3× value
Spurious drug	7 years → life	≥ ₹3 lakh or 3× value
Other contravention	1 → 2 years	≥ ₹20,000

Labelling and Packing of Drugs

The Rules prescribe:

- what information must appear on a drug label;
- where the information must appear;
- how the name of the drug should be displayed;
- declaration of quantity and active ingredients;
- manufacturer's details;
- batch number;
- manufacturing licence number;
- date of manufacture and expiry;
- special warnings for Schedule G, H, H1 and X drugs;
- requirements for external-use preparations;
- special requirements for veterinary preparations;
- special requirements for imported drugs;
- packing/pack-size requirements;
- special labelling requirements for cosmetics.

Labelling and Packing of Drugs Other than Homoeopathic Medicines

1. Rule 94: Exemption of Certain Drugs

1a. Drugs intended for export

Labels on packages or containers of drugs intended for export may be adapted to meet the specific requirements of the country to which the drug is exported.

Particular	Requirement
Name of drug	Must be displayed
Name and address of manufacturer	Must be displayed
Manufacturing licence number	Must be displayed
Batch/Lot number	Must be displayed
Date of expiry	Where applicable

The Rules also permit a **code number approved by the Licensing Authority** in certain circumstances where the consignee does not require the manufacturer's name and address.

1b. Prescription Medicines

Rules do not apply to a medicine which is made ready for treatment, with or without dilution; OR supplied on the prescription of a registered medical practitioner.

However, the medicine must still carry certain particulars.

The label must contain:

- Name and address of the supplier
- Name of the patient
- Quantity of medicine
- Serial number of the entry in the prescription register
- **Dose**, if the medicine is for internal use
- The words **“FOR EXTERNAL USE ONLY”**, if the medicine is for external application.

Rule 95 – Prohibition of Sale or Distribution Unless Labelled

No person shall sell or distribute any drug, including a patent or proprietary medicine, unless it is labelled in accordance with the Rules.

Under **Section 17**, a drug is considered **misbranded** if:

- it is not labelled in the prescribed manner; or
- its label/container/accompanying material contains a false or misleading statement, design or device.

Similarly, under **Section 17C**, a cosmetic is misbranded if it is not labelled in the prescribed manner or carries a false/misleading statement

Rule 96 – General/Manner of Labelling

The particulars must be:

- printed or written in indelible ink;
- displayed in a conspicuous manner;
- on the label of the innermost container;
- and on every other covering in which the container is packed.

General Labelling Requirements under Rule 96

1. Drug Name: The label must contain the **name of the drug**.

The **proper name of the drug must be printed/written more conspicuously than the trade name**, if a trade name exists.

The trade name should appear **immediately after or under the proper name**.

Category of drug	**Name to be used**
Drugs in **Schedule F/F(1)**	Name given in Schedule F/F(1)
Drugs in **Indian Pharmacopoeia/official pharmacopoeias**	Pharmacopoeial name/synonym followed by **I.P.** or appropriate recognized abbreviation
Drugs in **National Formulary of India**	Name/synonym given in NFI followed by **N.F.I.**
Other drugs	International Non-proprietary Name (INN), where available
If no INN exists	Name descriptive of the true nature or origin of the substance

2. Net Content: The label must contain a **correct statement of the net content**.

Depending upon the dosage form, this may be expressed in terms of weight, measure, volume, number of units & number of units of activity.

The **metric system** must be used for weight, measure and volume.

3. Declaration of Active Ingredients

The content of active ingredients must be stated in a manner appropriate to the dosage form.

For Example; Oral liquid preparations

- Active ingredients are expressed in terms of their content per **single dose**, normally indicated in **5 mL**.
- If the dose is below 5 mL, the content may be expressed per **1 mL or fraction thereof**.
- If the single dose is more than 5 mL, the content should be expressed in terms of the **minimum single dose approved by the Licensing Authority**.

For Liquid parenteral preparations

The active ingredient is expressed in terms of:

- amount per **1 mL**;
- percentage by volume; or
- amount per dose in the case of a single-dose container.

For an ampoule, the composition may be displayed on the label/wrapper of the package in which the ampoule is issued for sale.

Important exception: The requirement regarding declaration of active ingredient content does not apply in the same manner to pharmacopeial preparations where the composition is already specified in the relevant pharmacopoeia, and to preparations included in the National Formulary of India.

4. Name and Address of Manufacturer

The label must contain; **Name of the manufacturer and address of the premises where the drug has been manufactured.**

5. Batch Number: Every drug must have a **distinctive batch number**. The batch number is the number by reference to which the manufacturing details of that particular batch can be traced and made available for inspection.

6. Manufacturing Licence Number

7. Date of Manufacture and Expiry: Drugs specified in **Schedule P**, including their preparations and combinations, must bear, date of manufacture & date of expiry of potency. The period between manufacture and expiry must not exceed the period specified in Schedule P under the specified storage conditions.

Drugs not included in Schedule P must bear, date of manufacture & date of expiry. The expiry period should not exceed **60 months from the date of manufacture**, subject to possible extension by the Licensing Authority where satisfactory evidence is provided.

8. Physician Sample: Any drug intended for distribution to the medical profession as a **free sample** must comply with the normal labelling provisions. In addition, the label must be overprinted with: **“Physician's Sample—Not to be sold”**

9. Red Vertical Line Requirement: The line:

- must be on the **left side**;
- must run throughout the body of the label;
- must be **at least 1 mm in width**;
- must not disturb other information on the label.

Exceptions

- preparations intended for animal treatment;
- preparations intended for external use;
- ophthalmic preparations and ear drops;
- sterile preparations such as sutures, surgical dressings and preparations intended for parenteral use.

10. Imported Drugs: Drugs and their preparations, including combinations imported into India, must additionally bear:
Import Licence number, preceded by the words **"Import Licence"**; & **name and address of importer**.

11. Contraceptives: Mechanical contraceptives must carry the particulars specified in **Schedule R**. For contraceptives other than mechanical contraceptives, additional particulars include:
date of manufacture; date up to which the contraceptive is expected to retain its properties; & storage conditions required to preserve its properties.
For **oral contraceptives**, displaying the **date of manufacture only** is sufficient under this provision.

12. Labelling on Covers and Wrappers

The Rules do not require labelling of transparent covers; wrappers; cases; & other coverings
when they are used **solely for packing, transport or delivery**

13. Marking Directly on the Container

Where the Rules require information on a label, the information may instead be:

- etched;
- painted; or
- otherwise indelibly marked

on the container.
However, unless otherwise provided, the **name of the drug or distinctive letters referring to the drug should not be etched/painted/indelibly marked on glass containers other than ampoules**.
The expiry date is expressed in terms of **month and year** and indicates that the drug is recommended for use up to the **last day of that month**. The date must be preceded by:
"Expiry date"

Rule 97 – Labelling of Medicines

It is particularly important because it specifies the **special warnings and symbols** required for drugs belonging to certain Schedules.

1. Schedule G Drugs

A medicine for internal use containing a substance specified in **Schedule G** must bear, **"Caution: it is dangerous to take this preparation except under medical supervision".**

The statement must:

- be conspicuously printed;
- be surrounded by a line;
- have no other words within that line.

2. Schedule H Drugs

A medicine containing a substance specified in **Schedule H** must bear, **Symbol "Rx".**

The symbol must be conspicuously displayed at the **left top corner** of the label.

It must also carry **"Schedule H drug – Warning: To be sold by retail on the prescription of a Registered Medical Practitioner only"**

3. Schedule H Drugs Falling under NDPS Act

Where a Schedule H substance also falls under the **Narcotic Drugs and Psychotropic Substances Act, 1985**, the label must carry, **Symbol "NRx"**

The symbol must:

- be in **red**;
- be displayed conspicuously;
- appear at the **left top corner**.
- The label must also contain:
- **"Schedule H drug – Warning: To be sold by retail on the prescription of a Registered Medical Practitioner only".**

4. Schedule X Drugs

For a medicine containing a substance specified in **Schedule X**, the label must carry, **Symbol "XRx".**

The symbol must:

- be in **red**;
- be conspicuously displayed;
- be placed at the **left top corner**.
- The label must additionally state:
- **"Schedule X drug – Warning: To be sold by retail on the prescription of a Registered Medical Practitioner only"**

5. Schedule H1 Drugs

For a formulation containing a drug substance specified in **Schedule H1 must carry Symbol Rx**

The symbol must be **red**; conspicuously displayed; appear at the **left top corner**. The label must also carry the following warning **inside a box with a red border**:

"SCHEDULE H1 DRUG – WARNING. It is dangerous to take this preparation except in accordance with the medical advice. Not to be sold by retail without the prescription of a Registered Medical Practitioner."

6.Veterinary Medicines

A medicine prepared only for treatment of animals must be conspicuously labelled:

"Not for human use; for animal treatment only"

It must also bear a **symbol depicting the head of a domestic animal**.

7. Medicines Containing Industrial Methylated Spirit

A medicine prepared for treatment of human ailments that contains **industrial methylated spirit** must: indicate this fact on the label; and carry the words: **"For External Use Only"**

8. Medicines for Food-Producing Animals

A medicine intended for treatment of **food-producing animals** must indicate the **withdrawal period** for the species for which the drug is intended.

The withdrawal period is the interval between:

last administration of the veterinary medicine → production of food

*The concept of **withdrawal period** is important in veterinary pharmaceutical regulation because drug residues may enter food.*

Other Important Drug-labelling Provisions

1. Non-Sterile Surgical Ligature and Suture: Every container/wrapper enclosing a non-sterile surgical ligature or suture must bear, in a conspicuous manner and in indelible red ink, the words:

"Non-sterile surgical ligature (suture) – not to be used for operations upon the human body unless efficiently sterilized".

2. For a patent or proprietary medicine: Label must include Name and address of manufacturer, The true formula or list of ingredients must be printed or written in indelible ink on the outer label of every package containing the patent or proprietary medicine.

3. Use of "I.P." and Other Pharmacopeial Abbreviations: The letters, **I.P.** and recognized abbreviations of other pharmacopoeias/official compendia may be used on the label **only to indicate that the drug complies with the standards** laid down in that pharmacopoeia or official compendium.

4. Alteration of Labels: No person shall alter; obliterate; or deface any inscription or mark made or recorded by the manufacturer on the container; label; or wrapper of a drug.

Exception: alteration may be made at the **instance, direction or with permission of the Licensing Authority**.

Packing of Drugs

1. Pack Sizes: The pack sizes of drugs intended for retail sale must be as prescribed in **Schedule P1**. For drugs not covered by Schedule P1, specified pack sizes are prescribed.

Dosage form	Pack size requirement
Tablets/Capsules <10 units	Integral number
Tablets/Capsules >10 units	Multiples of 5
Liquid oral preparations	30 mL (paediatric only), 60, 100, 200, 450 mL
Paediatric oral drops	5, 10, 15 mL
Eye/Ear/Nasal drops	3, 5, 10 mL
Eye ointment	3, 5, 10 g

The above pack-size requirements do not apply to several categories, including:

1. Pack sizes/dosage forms not covered by the specified provisions;
2. imported formulations in finished form;
3. veterinary preparations;
4. preparations intended for export;
5. vitamins, tonics, cough preparations, antacids and laxatives in liquid oral unit-dose forms;
6. dosage forms intended for retail sale to hospitals, RMPs and nursing homes;
7. physician's samples;
8. large-volume intravenous fluids.

2. Packing of Schedule X Drugs

Schedule X drugs have restricted pack sizes.

They shall be marketed in packings not exceeding:

Dosage form	Maximum pack
Tablets/Capsules	**100 unit doses**
Oral liquid preparations	**300 mL**
Injections	**5 mL**

There is an exception for hospital/dispensary packs subject to specified conditions.

LABELLING OF COSMETICS

No person shall sell or distribute a cosmetic unless:

- an Indian-origin cosmetic is manufactured by a **licensed manufacturer**; and
- it is **labelled and packed according to the Rules**.
- Labels on packages/containers of cosmetics not manufactured for consumption or sale in India may be adapted according to the specific requirements of the consignee. If the consignee does not require the manufacturer's name and address, a **code number approved by the Licensing Authority** may be used.

Manner of Labelling Cosmetics

A cosmetic has requirements for:

- inner label;
- outer label;
- net contents;
- hazard information;
- batch number;
- manufacturing licence number;
- ingredient declaration;
- BIS-related requirements.

Both the inner and outer labels must carry:

1. Name of the cosmetic

2. Name and address of manufacturer

- The complete address of the premises where the cosmetic was manufactured should be given.
- For a very small container where the complete address cannot be accommodated:

Manufacturer's name; principal place of manufacture; PIN code should be given.

Net Contents on Outer Label

The **outer label** must carry a declaration of net contents.

Exceptions: The net-content declaration need not appear for:

- ***perfume/toilet water or similar package not exceeding 60 mL;***
- ***solid/semi-solid cosmetic package not exceeding 30 g.***

Cosmetic type	Declaration
Solid	Weight
Liquid	Fluid measure
Semi-solid	Weight
Sub-divided contents	Appropriate numerical count along with quantity

Hazard Information – Inner Label

Where a hazard exists, the inner label must contain:

1. Adequate directions for safe use
2. Warnings/cautions/special directions to be observed by the consumer
3. Names and quantities of hazardous or poisonous ingredients

Cosmetic Batch Number

The batch number must be preceded by the letter, "**B".** For example; **B 12345.**

However, this requirement does not apply to cosmetics containing, **10 g or less**, if solid/semi-solid; or **25 mL or less**, if liquid.

For soaps, instead of batch number, the **month and year of manufacture** must be given on the label.

Manufacturing Licence Number for Cosmetics

The manufacturing licence number must be preceded by "**M"**

For **Example M 12345.**

Declaration of Ingredients in Cosmetics

The label must carry the heading; **"INGREDIENTS"**

- Ingredients present in concentrations **more than 1%** must be listed in descending order of weight or volume at the time they are added.
- Ingredients present at concentrations **1% or less** may then be listed in any order.

Exception

The ingredient statement need not appear for packs containing less than:

- **60 mL of liquid**, or
- **30 g of solid/semi-solid**.

Hair Dyes – Special Labelling

Rule 149 provides special labelling requirements for hair dyes containing:

- Para-phenylenediamine;
- other dyes;
- colours;
- pigments.

The warning must appear on **both inner and outer labels**, in:

- English; and
- local languages.

The label must carry a caution concerning possible skin irritation and the requirement for a **preliminary test**.

It must also warn that the preparation should **not be used for dyeing eyelashes or eyebrows**, because such use may cause severe eye inflammation or blindness

Toothpaste Containing Fluoride

Rule 149A provides special requirements for toothpaste containing fluoride.

Fluoride content must not exceed:**1000 ppm**

The fluoride content in ppm and date of expiry must be mentioned on tube; & carton.

LIST OF COLOURS PERMITTED & OFFENCES AND PENALTIES

A manufacturer cannot use an arbitrary colouring agent in a pharmaceutical preparation. Only the colours specified in **Rule 127** are permitted.

The colours are broadly divided into:

1. Natural colours (chloroform, curcumin, carotene, Annatto, Cochineal, Titanium Oxide, Red oxide of Iron etc)
2. Artificial colours (Caramel, Riboflavin)
3. Coal tar colours
4. Lakes

Coal Tar Colours

The coal tar colours are given with:

- **Common name**
- **Colour Index Number**
- **Chemical name**

This is an important part of Rule 127 because the permitted colours are identified not merely by their common names but also by their Colour Index numbers and chemical descriptions.

Common name	Colour Index No.	Chemical name
Quinazarine Green S.S.	61565	1,4-bis (p-Toluino) anthraquinone
Alizarin Cyanine Green F.	61570	Disodium salt of 1,4-bis (O-sulfo-p-Toluino) anthraquinone
Fast Green F.C.F.	42053	Disodium salt of the specified sulfonium-substituted anthraquinone compound
Tartrazine	19140	Trisodium salt of 3-carboxy-5-hydroxy-1-p-sulfophenyl-4-p-sulfophenylazopyrazole
Sunset Yellow FCF	15985	Disodium salt of 1-p-sulfophenyl azo-2-naphthol-6-sulfonic acid

Lakes

Rule 127 also permits, **the aluminium or calcium salts (lakes) of any of the water-soluble colours listed above.**

Thus, lakes prepared from the permitted water-soluble colours can also be used.

What is a lake?

A **lake colour** is an insoluble pigment produced by converting a water-soluble dye into an insoluble form, commonly using a metallic salt/substrate such as aluminium or calcium.

This makes the colour more suitable for certain dosage forms where a water-soluble dye would not be appropriate.

Cosmetics: Permitted colours

Rule 134: No cosmetic shall contain dyes, colours or pigments other than those specified by:

Bureau of Indian Standards: IS 4707 Part I, as amended, and **Schedule Q**.

The permitted synthetic organic and natural organic colours used in cosmetics are also subject to limits for certain contaminants.

Substance	Maximum permitted level
Arsenic, calculated as arsenic trioxide	2 ppm
Lead, calculated as lead	20 ppm
Heavy metals other than lead	100 ppm


THE END